HW 1

1. Denote the population by a random variable (r.v.) $X \sim B(1,p)$, i.e., $P(X=1)=p,\ P(X=0)=1-p$, where p is an unknown parameter. Let $\boldsymbol{X}=(X_1,X_2,\cdots,X_6)$ be a random sample from X,
   - Write out the sample space and the probability distribution of $\boldsymbol{X}$;
   - Point out which of the followings are statistics: $X_1,\ X_1-X_2, X_{(1)},\ X_5+p^2,\ X_4-E(X_1),\ (X_5-X_1)^2/\mathrm{Var}(X_6)$.

- sample space $\mathcal{X}=\{(x_1,x_2,x_3,\cdots x_6):\ x_i=1,0\}$

$$P(X_1=x_1,X_2=x_2,X_3=x_3,\cdots X_6=x_6)=\begin{cases} p^a(1-p)^{6-a}, & x_i=0,1,\ \sum_{i=1}^{6}x_i=a \\ 0, & \text{else}\end{cases}$$

- statistics: X_1, X_1-X_2, $X_{(1)}$. for p is unknown

2. The independent r.v. X and Y are distributed as $B(m,p)$ and $B(n,p)$ (binomial distribution), respectively.
   - What is the distribution of the r.v. $X+Y$?
   - If $m=8, n=12$, and $p=0.25$, what is the numerical value of the probability: $P(10\le X+Y\le 15)$?

- $X+Y \sim B(m+n,p)$
- $X+Y \sim b(20,\frac{1}{4})$ $\quad P(10\le X+Y\le 15)=\sum_{k=10}^{15}\binom{20}{k}(\frac{1}{4})^k(\frac{3}{4})^{20-k}\approx 0.01386$

3. Let $\boldsymbol{X}=(X_1,X_2,\cdots,X_n)$ be a random sample from the population $X\sim N(\mu,0.5)$, determine the sample size n such that $P(|\bar{X}-\mu|<0.1)\ge 0.997$.

$$P(|\bar{X}-\mu|<0.1)=2\Phi\left(\frac{0.1}{\sqrt{0.5/n}}\right)-1\ge 0.997$$

$$\Rightarrow n=444$$

4. Let $X_1,\cdots,X_n$, $n=10$, be a random sample from a population with an unknown distribution F and their observed values are

$$0.2, 0.1, -0.15, 0.2, -0.2, 1, 0.18, -1.2, 0.5, -1.2.$$

Let $F_n(x)$ be the empirical distribution function. Give an approximation for estimation error: $P(|F_n(0)-F(0)|\ge 0.01)$ (for simplicity, suppose that you can use laws of large numbers and central limit theory).

$-1.2\ \ -1.2\ \ -0.2\ \ -0.15\ \ 0.1\ \ 0.18\ \ 0.2\ \ 0.2\ \ 0.5\ \ 1$

$F_n(0)=\frac{4}{10}$ $\quad \frac{\sqrt{n}(F_n(x)-F(x))}{\sqrt{F(x)(1-F(x))}}\xrightarrow{d}N(0,1)$ $\quad F(0)\xrightarrow{P}F_n(0)=\frac{2}{5}$

$$P(|F_n(0)-F(0)|\ge 0.01)=2-2\Phi\left(\frac{0.01}{\sqrt{\frac{F(0)(1-F(0))}{n}}}\right)=2-2\Phi(0.085)\approx 0.9486$$

5. Let $X_1, \ldots, X_n$ *i.i.d.* $\sim N(\mu, \sigma^2)$, $\bar{X} = n^{-1}\sum_{i=1}^n X_i$, and $S_n^2 = n^{-1}\sum_{i=1}^n (X_i - \bar{X})^2$. Let $X_{n+1} \sim N(\mu, \sigma^2)$ and suppose that X_{n+1} is independent of $X_1, \ldots, X_n$. Determine the distribution of $(X_{n+1} - \bar{X})/S_n$.

$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu, \frac{\sigma^2}{n})$ $\quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ $\quad \therefore \frac{(n-1)S^2}{\sigma^2} = \frac{nS_n^2}{\sigma^2} \sim \chi^2_{n-1}$

$X_{n+1} - \bar{X} \sim N(0, \frac{\sigma^2}{n} + \sigma^2)$ $\quad \therefore \frac{X_{n+1} - \bar{X}}{\sigma\sqrt{\frac{1}{n}+1}} \sim N(0,1)$ $\quad \therefore \frac{X_{n+1} - \bar{X}}{S_n} \cdot \sqrt{\frac{n-1}{n+1}} \sim t_{n-1}$

$\therefore \frac{X_{n+1} - \bar{X}}{S_n} \sim \sqrt{\frac{n+1}{n-1}}\, t_{n-1}$

6. Let $X \sim \chi^2_n$.

- Show that $E(X) = n$, $\mathrm{Var}(X) = 2n$;
- Calculate the population kurtosis $\beta_2 = \{E(X - EX)^4\}/\{E(X - EX)^2\}^2 - 3$;
- Show that $(X - n)/\sqrt{2n} \xrightarrow{d} N(0,1)$ as $n \to \infty$;
- If $X_i \sim \chi^2_{n_i}$, $i = 1, 2, \cdots, k$ and $X_1, \cdots, X_k$ are independent, let $m = \sum_{i=1}^k n_i$, then using characteristic function method to show that $\sum_{i=1}^k X_i \sim \chi^2_m$.

- $X = \sum_{i=1}^{n} X_i^2$ $\quad X_i \sim N(0,1)$ i.i.d $\quad EX_i^2 = 1$ $\quad \mathrm{var}\, X_i^2 = EX_i^4 - (EX_i^2)^2 = 3 - 1 = 2$

$\therefore EX = n, \quad \mathrm{var}\, X = 2n$

- mgf of X_n^2 $\quad M_n(s) = (1-2s)^{-\frac{n}{2}} \xrightarrow{\text{Taylor}} 1 + ns + \frac{1}{2}n(n+2)s^2 + \frac{1}{6}n(n+2)(n+4)s^3 + \frac{1}{24}n(n+2)(n+4)(n+6) + \cdots$

$\therefore EX^2 = n(n+2)$ $\quad EX^3 = n(n+2)(n+4)$ $\quad EX^4 = n(n+2)(n+4)(n+6)$

$$\beta_2 = \frac{E(X-n)^4}{(E(X-n)^2)^2} - 3 = \frac{12}{n}$$

- $EX = n, \quad \mathrm{var}\, X = 2n$

considering CLT, for n given $EX_1, \mathrm{var}\, X_1 < \infty$

$$\frac{X - n}{\sqrt{2n}} \xrightarrow{d} N(0,1)$$

- the cf of χ^2_n $\quad \phi_n(t) = (1-2is)^{-\frac{n}{2}}$

due to the independence of X_i

$$\phi_{\sum_{i=1}^k X_i}(t) = (1-2is)^{-\frac{n_1}{2}} \cdot (1-2is)^{-\frac{n_2}{2}} \cdot (1-2is)^{-\frac{n_3}{2}} \cdots (1-2is)^{-\frac{n_k}{2}} = (1-2is)^{-\frac{1}{2}\sum_{i=1}^k n_i}$$

$$= (1-2is)^{-\frac{m}{2}}$$

$$\Rightarrow \sum_{i=1}^{k} X_i \sim \chi^2_m$$

7. Let $X \sim \Gamma(p, \alpha)$ (Gamma distribution).

   - Show that $E(X) = p/\alpha$, $\mathrm{Var}(X) = p/\alpha^2$;
   - If $X_i \sim \Gamma(p_i, \alpha)$, $i = 1, 2, \cdots, n$, and $X_1, \cdots, X_n$ are independent, let $p^* = \sum_{i=1}^n p_i$, then $\sum_{i=1}^n X_i \sim \Gamma(p^*, \alpha)$.

the PDF of Gamma distribution: $f(x) = \frac{\alpha^p}{\Gamma(p)} x^{p-1} e^{-\alpha x} 1_{x>0}$

- $EX = \int_0^\infty \frac{\alpha^p}{\Gamma(p)} x^p e^{-\alpha x} dx = \frac{p}{\alpha} \int_0^\infty \frac{\alpha^{p+1}}{\Gamma(p+1)} x^p e^{-\alpha x} dx = \frac{p}{\alpha}$

  $EX^2 = \frac{p(p+1)}{\alpha^2} \int_0^\infty \frac{\alpha^{p+2}}{\Gamma(p+2)} x^{p+1} e^{-\lambda x} dx = \frac{p(p+1)}{\alpha^2} \Rightarrow \mathrm{var}\, X = EX^2 - (EX)^2 = \frac{p}{\alpha^2}$

- considering the cf of Gamma distribution $\phi(t) = \left(\frac{\alpha}{\alpha - it}\right)^p$

  due to the independence of X_i

  $\phi_{\sum_{i=1}^n X_i}(t) = \left(\frac{\alpha}{\alpha - it}\right)^{p_1} \left(\frac{\alpha}{\alpha - it}\right)^{p_2} \cdots = \left(\frac{\alpha}{\alpha - it}\right)^{p^*}$

  $\Rightarrow \sum_{i=1}^n X_i \sim \Gamma(\alpha, p^*)$

8. The p.d.f. of the Weibull distribution with two unknown parameters α and β is

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, \quad x > 0 \quad (\alpha, \beta > 0).$$

Let $X_1, \ldots, X_n$ be a random sample from this distribution. Let $Y = X_{(1)}$ be the smallest order statistic.

   - Is the family of Weibull distributions an exponential family?
   - Obtain the p.d.f. of Y.
   - Calculate the expectation $E(Y)$ and variance $\mathrm{Var}(Y)$.

- $f(x) = \alpha\beta \cdot e^{(\beta-1)\ln x} e^{-\alpha} e^{\beta \ln x}$ — $q(x)$ depend on parameter β

  No

- $F(x) = \int_0^x \alpha\beta s^{\beta-1} e^{-\alpha s^\beta} ds = 1 - e^{-\alpha x^\beta}$

  $Y: f(x_{(1)}) = \frac{n!}{(n-1)!} f(x) [1 - F(x)]^{n-1} = n\alpha\beta x^{\beta-1} e^{-n\alpha x^\beta}$

  $\therefore Y \sim \mathrm{Weibull}(n\alpha, \beta)$

- consider the EX of $X \sim \mathrm{Weibull}(\alpha, \beta)$

  $EX = \int_0^\infty \alpha\beta x^\beta e^{-\alpha x^\beta} dx = \int_0^\infty x e^{-\alpha x^\beta} d\alpha x^\beta = \alpha^{-\frac{1}{\beta}} \Gamma(1 + \frac{1}{\beta})$

  $\mathrm{var}\, X = \alpha^{-\frac{2}{\beta}} \left[\Gamma(1 + \frac{2}{\beta}) - \Gamma^2(1 + \frac{1}{\beta})\right]$

  $\therefore EY = (n\alpha)^{-\frac{1}{\beta}} \Gamma(1 + \frac{1}{\beta})$ $\quad \mathrm{var}\, Y = (n\alpha)^{-\frac{2}{\beta}} \left[\Gamma(1 + \frac{2}{\beta}) - \Gamma^2(1 + \frac{1}{\beta})\right]$

HW 2

1. Let $X_1, \cdots, X_n$ be a random sample from the distribution with p.d.f.

$$f(x;\theta) = \frac{1}{2\theta}\exp\left\{-\frac{|x|}{\theta}\right\}, \quad -\infty < x < +\infty,\ \theta > 0.$$

Show that $T = \sum_{i=1}^n |X_i|$ is a sufficient and complete statistic of θ.

$f(\vec{x};\theta) = \prod_{i=1}^n \frac{1}{2\theta}\exp\left\{-\frac{|x_i|}{\theta}\right\} = \left(\frac{1}{2\theta}\right)^n e^{-\frac{1}{\theta}\sum_{i=1}^n x_i} = \left(\frac{1}{2\theta}\right)^n e^{-\frac{1}{\theta}t} = g(t(x),\theta)\cdot 1$

By factorization theorem, T is a sufficient statistic of θ

$f(\vec{x};\theta)$ is in an exponential family

$\Theta^* = \{\varphi = -\frac{1}{\theta} \mid \varphi < 0\}$ as a subset of R^1 has interior points

T is a complete statistic of θ

2. Let $X_1, \cdots, X_m$ *i.i.d.* $\sim N(a,\sigma^2)$, $Y_1, \cdots, Y_n$ *i.i.d.* $\sim N(b,\sigma^2)$, and X_i's and Y_j's are independent. Derive the sufficient and complete statistic of (a, b, σ^2).

Let $\bar{X}$ be $\frac{1}{m}\sum_{i=1}^m X_i$, $\bar{Y} = \frac{1}{n}\sum_{i=1}^n Y_i$ $\quad S^2 = \frac{1}{m+n}\left[\sum_{i=1}^m (X_i-\bar{X})^2 + \sum_{i=1}^n (Y_i-\bar{Y})^2\right]$

due to the independence, $f(\vec{X},\vec{Y};a,b,\sigma^2) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{m+n} e^{\left(-\frac{1}{2\sigma^2}\sum_{i=1}^m (x_i-a)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-b)^2\right)}$

$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{m+n} e^{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^m x_i^2 - 2a\sum_{i=1}^m x_i + ma^2 + \sum_{i=1}^n y_i^2 - 2b\sum_{i=1}^n y_i + nb^2\right)}$

$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{m+n} e^{-\frac{ma^2-nb^2}{\sigma^2} + \frac{1}{\sigma^2}\left(\sum_{i=1}^m x_i a + \sum_{i=1}^n y_i b\right) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^m x_i^2 + \sum_{i=1}^n y_i^2\right)}$

$= f(t(x),\theta)\cdot h(x)$, where $h(x) \equiv 1$

By factorization theorem, $\bar{X}, \bar{Y}, S^2$ is sufficient statistic of (a, b, σ^2)

since $f(\vec{X},\vec{Y};a,b,\sigma^2)$ is in exponential family

$\Theta^* = \left\{\frac{a}{\sigma^2}, \frac{b}{\sigma^2}, -\frac{1}{2\sigma^2} \mid -\infty < \frac{a}{\sigma^2} < +\infty,\ -\infty < \frac{b}{\sigma^2} < +\infty,\ -\frac{1}{2\sigma^2} < 0\right\}$ has interior points

$\therefore \bar{X}, \bar{Y}, S^2$ is complete statistic of (a, b, σ^2)

3. Let $X_1, \cdots, X_n$ *i.i.d.* $\sim U(-\theta/2, \theta/2)$, $\theta > 0$, show that $(X_{(1)}, X_{(n)})$ is sufficient but not complete.

no nonconstant function of T is ancillary

$f(\vec{x};\theta) = \frac{1}{\theta^n}\prod_{i=1}^n I_{-\frac{\theta}{2} < x_i < \frac{\theta}{2}} = \frac{1}{\theta^n}\cdot I_{-\frac{\theta}{2} < x_{(1)} \le x_{(n)} < \frac{\theta}{2}} = g(t(x),\theta)\cdot h(x),\ h(x) \equiv 1$

By factorization theorem, $X_{(1)}, X_{(n)}$ is sufficient

Let $Y_i = \theta X_i + \frac{1}{2} \sim U(0,1)$ i.i.d $\quad Z = Z(\vec{Y}) = X_{(n)} - X_{(1)} = Y_{(n)} - Y_{(1)}$ 与θ无关

there exist constants a, b such that $P(Z < a) = P(Z > b) > 0$

$\varphi(t) = \begin{cases} 1 & Z(t) < a \\ -1 & Z(t) > b \\ 0 & \text{else} \end{cases}$ $\quad E_\theta\varphi(t) = 0$ but $P(\varphi(t) = 0) > 0$ $\quad \therefore (X_{(1)}, X_{(n)})$ is not complete

4. Let $X_1, \cdots, X_n$ be a random sample from two parameter exponential distribution with p.d.f.

$$f(x;\lambda,\mu) = \lambda^{-1} \exp\left\{-\frac{x-\mu}{\lambda}\right\} I_{\{x>\mu\}},$$

where $0 < \lambda < +\infty$, $-\infty < \mu < +\infty$ are two unknown parameters. Show that

- $\left(X_{(1)}, \sum_{i=1}^n X_{(i)}\right)$ is sufficient for (λ, μ);
- $X_{(1)}$ is independent of $\sum_{i=1}^n (X_i - X_{(1)})$.

- since X_i are independent, $f(\vec{x};\lambda,\mu) = \lambda^{-n} \exp\left\{-\frac{\sum_{i=1}^n x_i - n\mu}{\lambda}\right\} \prod_{i=1}^n I_{x_i>\mu}$

$= \lambda^{-n} \exp\left\{-\frac{\sum_{i=1}^n (x_{(i)}-\mu)}{\lambda}\right\} \cdot I_{x_{(1)}>\mu} = g(t(\vec{x}),\theta) \cdot h(\vec{x})$, where $h(\vec{x}) \equiv 1$

By factorization theorem, $(X_{(1)}, \sum_{i=1}^n X_{(i)})$ is sufficient for (λ, μ)

- fix λ: $f(\vec{x};\mu) = \lambda^{-n} \exp\left\{-\frac{\sum_{i=1}^n (x_i-\mu)}{\lambda}\right\} I_{x_{(1)}>\mu} = g(t(\vec{x}),\mu) \cdot h(\vec{x})$

$\therefore$ $X_{(1)}$ is sufficient statistic of μ

$f(x_{(1)}) = n f(x) (1-F(x))^{n-1} = n \frac{1}{\lambda} e^{-\frac{x-\mu}{\lambda}} n \cdot I_{x>\mu}$

Let $t = X_{(1)}$ and $\varphi(t)$ be a function satisfying $E_\mu \varphi(t) = 0$

$\frac{n}{\lambda} \int_\mu^{+\infty} \varphi(t)\, e^{-\frac{n}{\lambda}(t-\mu)} dt = 0 \quad \therefore \int_\mu^{+\infty} \varphi(t)\, e^{-\frac{n}{\lambda}t} dt = 0$

Differentiate with respect to μ $\quad \varphi(\mu) e^{-\frac{n}{\lambda}\mu} = 0 \Rightarrow \varphi(\mu) = 0 \quad \forall \mu \in R$

$\therefore$ $X_{(1)}$ is complete statistic of μ

$Y_i = X_i - \mu \quad f(y) = \frac{1}{\lambda} e^{-\frac{y}{\lambda}} I_{y>0}$ $Y_i - Y_{(1)} = X_i - X_{(1)}$ does not depend on μ

By Basu theorem, $X_{(1)}$ is independent of $\sum_{i=1}^n (X_i - X_{(1)})$

5. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from exponential distribution with p.d.f.

$$f(x;\theta) = \exp\{-(x-\theta)\}\, I_{\{x>\theta\}}, \quad -\infty < \theta < +\infty.$$

Derive the moment estimator of θ and show that it is unbiased.

$\alpha_1 = E_\theta X_1 = \int_\theta^{+\infty} x e^{-x+\theta} dx = \theta + 1$ $\quad \therefore$ the moment estimator of θ is $\frac{1}{n}\sum X_i - 1$

$\hat{\theta} = \bar{X} = \alpha_{n1} \quad E\bar{X} = E\frac{1}{n}\sum_{i=1}^n X_i = \theta + 1$

$\therefore$ it is unbiased

6. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from exponential distribution with p.d.f.

$$f(x;\theta) = \theta e^{-\theta x},\ x > 0,\ \theta \in \Theta = (0, \infty).$$

- Show that both $\bar{X}$ and $nX_{(1)}$ are unbiased estimators of $1/\theta$;
- Which of these two estimators would you prefer?

— $EX_1 = \int_0^\infty x\theta e^{-\theta x} dx = \frac{1}{\theta}$ $\quad E\bar{X} = E(\frac{1}{n}\sum_{i=1}^n X_i) = E(\frac{n}{n\theta}) = \frac{1}{\theta}$

$E\, n X_{(1)} = n\, E X_{(1)} = n \int_0^\infty x\, n\theta e^{-n\theta x} dx = n \cdot \frac{1}{n\theta} = \frac{1}{\theta}$

$\therefore$ they are unbiased estimators of $1/\theta$

— $\bar{X}$ considering the variance of statistic considering consistence is ok

$\text{var}\, X_1 = \frac{1}{\theta^2}$ $\quad \text{var}\, \bar{X} = \text{var}(\frac{1}{n}\sum_{i=1}^n X_i) = \frac{1}{n\theta^2}$ $\quad X_{(1)} \sim \xi(n\theta)$

$\text{var}\, n X_{(1)} = n^2 \cdot \frac{1}{n^2\theta^2} = \frac{1}{\theta^2} > \text{var}\, \bar{X}$ $\quad \therefore$ $\bar{X}$ is better estimator

7. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Gamma distribution with parameters α, β and p.d.f.

$$f(x;\alpha,\beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0,\ \alpha > 0,\ \beta > 0.$$

Derive the moment estimators of α and β.

$EX_1 = \int_0^\infty \frac{x^\alpha}{\Gamma(\alpha)} \beta^{-\alpha} e^{-\frac{x}{\beta}} dx = \alpha\beta$ $\quad EX_1^2 = \int_0^\infty \frac{x^{\alpha+1}}{\Gamma(\alpha)} \beta^{-\alpha} e^{-\frac{x}{\beta}} dx = \alpha(\alpha+1)\beta^2$ $\quad \text{var}\, X_1 = \alpha\beta^2$

$\therefore$ $\beta = \frac{\text{var}\, X_1}{EX_1}$ $\quad \alpha = \frac{(EX_1)^2}{\text{var}\, X_1}$

considering moment estimators $\quad \bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ $\quad S_n^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$

$\hat{\beta} = \frac{S_n^2}{\bar{X}}$ $\quad \alpha = \frac{(\bar{X})^2}{S_n^2}$

8. Let $(X_1, Y_1), \cdots, (X_n, Y_n)$ be a random sample from the Hong Kong Children height data. Let $(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ be the population means, variances, and correlation of height and weight. The observed values of the sample with sample size $n = 200$ are given in the attached file "Homework_HongKong_Height_Weight.csv".

- Compute the moment estimators of $(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ (denoted as $(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2, \hat{\rho})$).
- Estimate the biases and variances of $(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2)$ (suppose X_i and Y_i are normally distributed).
- How to decrease the MSE of the above estimators?

— By computing in Excel

$\hat{\mu}_1 = 68.178$ $\quad \hat{\mu}_2 = 127.345$ $\quad \hat{\sigma}_1^2 = 4.149$ $\quad \hat{\sigma}_2^2 = 134.561$ $\quad \hat{\rho} = 0.432712$

— $E\hat{\mu}_1 - E\mu_1 = 0$ $\quad E\hat{\mu}_2 - E\mu_2 = 0$

$E\hat{\sigma}_1^2 = \frac{1}{n} E\sum_{i=1}^n (X_i - \mu)^2 = \frac{n-1}{n}\sigma_1^2$ $\quad E\hat{\sigma}_2^2 = \frac{n-1}{n}\sigma_2^2$

biases : $\quad E\hat{\sigma}_1^2 - \sigma_1^2 = -\frac{1}{n}\sigma_1^2$ $\quad E\hat{\sigma}_2^2 - \sigma_2^2 = -\frac{1}{n}\sigma_2^2$

$\text{var}(\hat{\mu}_1) = \text{var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sigma_1^2$ $\quad \text{var}(\hat{\mu}_2) = \frac{1}{n}\sigma_2^2$ $\quad \text{var}\, S^2 = \frac{2\sigma^4}{n-1}$

$\text{var}(\hat{\sigma}_1^2) = \text{var}\left(\frac{n-1}{n} S_n^2\right) = \frac{2(n-1)\sigma_1^4}{n^2}$

$\text{var}(\hat{\sigma}_2^2) = \frac{2(n-1)\sigma_2^4}{n^2}$ $\quad$ Calculate the numerical results

- To decrease the MSE, we need to increase n, such that bias & variance will drop.

since the MSE of S_n^2 is less than $\frac{1}{n-1}\sum_{i}^{n}(x_i - \bar{x})^2$

HW 3

1. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from the distribution with p.d.f.

$$f(x; \mu, \sigma) = \frac{1}{\sigma} \exp\Big\{ - \frac{x - \mu}{\sigma} \Big\}, \quad x \geq \mu.$$

   - When μ is known, derive the moment estimator and MLE of σ;
   - When σ is known, derive the moment estimator and MLE of μ;
   - When both μ and σ are unknown, derive the moment estimators and MLEs of μ, σ and $P(X_1 < t)$ ($t > \mu$ and t is known).

$EX_1 = \int_\mu^{+\infty} \frac{x}{\sigma} e^{-\frac{x-\mu}{\sigma}} dx = \mu + \sigma$ $\quad \text{var} X_1 = \sigma^2$

$L = \frac{1}{\sigma^n} e^{-\frac{x_1-\mu}{\sigma} - \frac{x_2-\mu}{\sigma} - \cdots}$ $\quad l = -n \ln\sigma - \frac{\sum x_i - n\mu}{\sigma}$

L is in exponential family

- when μ is known, the moment estimator can be $\bar{X} - \mu$

$\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \frac{\sum x_i - n\mu}{\sigma^2} = 0 \Rightarrow \sigma = \frac{1}{n}\sum_{i=1}^{n} x_i - \mu = \bar{X} - \mu$

- when σ is known, the moment estimator can be $\bar{X} - \sigma$

$\frac{\partial l}{\partial \mu} = \frac{n}{\sigma} > 0$, the likehood function ↑

since $\mu \leq X_{(1)} \leq X_{(2)} < \cdots$ the MLE of μ is $X_{(1)}$

- the moment estimator : $\hat{\mu} = \bar{X} - S_n$ $\quad \sigma = S_n$

  MLE : $\hat{\mu} = X_{(1)}$ $\quad \sigma = \bar{X} - X_{(1)}$

$P(X_1 < t) = \int_\mu^t \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}} dx = 1 - e^{-\frac{t-\mu}{\sigma}}$

So the moment estimator of $P(X_1 < t)$ is $1 - e^{-\frac{t - \bar{X} + S_n}{S_n}}$

MLE $\quad 1 - e^{-\frac{t - X_{(1)}}{\bar{X} - X_{(1)}}}$ (invariance property)

2. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from uniform distribution $U(\theta, 4\theta)$, $0 < \theta < +\infty$.

   - Derive the MLE of θ;
   - Is the MLE unbiased? If not, find an unbiased estimate based on the MLE;
   - Is the MLE weakly consistent? Why?

$f(x_i) = \frac{1}{3\theta} I_{\theta \leq x_i \leq 4\theta}$ $\quad L(\theta; \vec{x}) = (3\theta)^{-n} \cdot I_{\theta \leq x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)} \leq 4\theta}$

- the maxium value of the likehood function is obtained when $\frac{1}{4} x_{(n)} \leq \theta \leq x_{(1)}$

  so the MLE of θ is $\frac{1}{4} x_{(n)}$

- $f(x_{(n)}) = n \left(\frac{x - \theta}{3\theta}\right)^{n-1} \cdot \frac{1}{3\theta} I_{\theta \leq x \leq 4\theta}$

$E X_{(n)} = \int_\theta^{4\theta} x n \frac{(x-\theta)^{n-1}}{(3\theta)^n} dx = \frac{4n+1}{n+1}\theta \neq \theta$ so it is unbiased

considering $\frac{n+1}{4n+1} X_{(n)}$ is unbiased estimator of θ

$\forall \theta,\ \forall \varepsilon < \theta$

$P_\theta(|\frac{1}{4} X_{(n)} - \theta| > \varepsilon) = P_\theta(X_{(n)} > 4(\varepsilon+\theta) \text{ or } X_{(n)} < 4(\theta-\varepsilon))$

$= \int_{4(\theta-\varepsilon)}^{4\theta} n x \frac{(x-\theta)^{n-1}}{(3\theta)^n} dx = (1 - \frac{4\varepsilon}{4\theta})^n \to 0$

$\lim_{n\to\infty} P_\theta(|\frac{1}{4} X_{(n)} - \theta| > \varepsilon) = 0$

so, it is weakly consistent

3. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from the Weibull distribution with p.d.f.

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, \quad x > 0 \quad (\alpha, \beta > 0).$$

Suppose β is known, determine the MLE of α.

Likehood function: $L(\alpha, \vec{x}) = (\alpha\beta)^n (x_1 x_2 \cdots x_n)^{\beta-1} e^{-\alpha(x_1^\beta + x_2^\beta + \cdots x_n^\beta)}$

is in exponential family

$l(\alpha, \vec{x}) = n \ln \alpha\beta + (\beta-1) \sum_{i=1}^n \ln x_i - \alpha \sum_{i=1}^n x_i^\beta$

$\frac{\partial l}{\partial \alpha} = \frac{n}{\alpha} - \sum_{i=1}^n x_i^\beta = 0 \Rightarrow \alpha = \frac{n}{\sum_{i=1}^n x_i^\beta}$

4. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Geometric distribution:

$$P(X_1 = i) = \theta(1-\theta)^{i-1}, \ i = 1, 2, \cdots, \ 0 < \theta < 1.$$

Derive the UMVUE of θ^{-1} and θ.

$E X_i = \theta^{-1}$ $\quad$ $\text{Var } X_i = \frac{1-\theta}{\theta^2}$

$p(x, \theta) = P_\theta(X = x) = \theta(1-\theta)^{x-1} = \theta \cdot e^{(x-1)\ln(1-\theta)}$ is in exponential family

so is $P(\vec{x}, \theta)$ $\quad$ $\ln(1-\theta) \in (-\infty, 0)$ has interior points

$T(X) = \sum_{i=1}^n X_i$ is sufficient & complete statistic of θ

$E \frac{1}{n} T(x) = \theta^{-1}$ is unbiased of θ^{-1}, so the UMVUE of θ^{-1} is $\frac{1}{n} \sum_{i=1}^n X_i$

further, $P_\theta(T = t) = C_{t-1}^{n-1} \theta^n (1-\theta)^{t-n}$, $t = n+1, n+2 \cdots$

noticing $P(X_i = 1) = \theta$ $\quad$ Let $\varphi(X_i) = \begin{cases} 1, & X_i = 1 \\ 0, & \text{else} \end{cases}$

$E(\varphi(X_1) | T = t) = \frac{P(X_1 = 1, \sum_{i=2}^n X_i = t-1)}{P(T = t)} = \frac{\theta \cdot C_{t-2}^{n-2} \theta^{n-1} (1-\theta)^{t-n}}{C_{t-1}^{n-1} \theta^n (1-\theta)^{t-n}} = \frac{n-1}{t-1}$

so, the UMVUE of θ is $\frac{n-1}{T-1}$

5. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu_0, \sigma^2)$, where μ_0 is known and $\sigma > 0$ is unknown.

   - Derive the moment estimator and MLE of σ^2;
   - Derive the C-R lower bound for the variance of the unbiased estimator of σ^2;
   - Derive the UMVUE of σ^2;
   - Derive the MLE and UMVUE of σ.

- $\operatorname{var} X_i = \sigma^2$ $\quad S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i-\mu_0)^2$ is the moment estimator of σ^2

  Likelihood function $L(\sigma^2, \vec{X}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu_0)^2\right\}$

  $l = -\frac{n}{2}\ln 2\pi\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu_0)^2$

  $\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2}\frac{1}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i-\mu_0)^2 = 0 \Rightarrow \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i-\mu_0)^2$

  since the pdf is in exponential family, the MLE of σ^2 is S^2

- Let $\theta = \sigma^2$ $\quad f(X,\theta) = \frac{1}{\sqrt{2\pi\theta}}\exp\left\{-\frac{1}{2\theta}(X-\mu_0)^2\right\}$

  since $f(\vec{X},\theta)$ belongs to exponential family

  $I(\theta) = E\left(\frac{\partial \ln f}{\partial\theta}\right)^2 = \frac{1}{4\theta^4}E\left[(X-\mu_0)^2-\theta\right]^2 = \frac{1}{4\theta^4}\operatorname{var}(X-\mu_0)^2 = \frac{1}{2\theta^2}$

  $g(\theta) = \theta$ $\quad \therefore$ the C-R lower bound of σ^2 is $\frac{2\sigma^4}{n}$

- $T(X) = \sum_{i=1}^{n}(X_i-\mu_0)^2$ is sufficient & complete statistic of σ^2

  $E\left(\frac{1}{n}T(X)\right) = \frac{1}{n}\cdot n\,E(X_i-\mu_0)^2 = \sigma^2$

  $\therefore$ $\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu_0)^2$ is UMVUE of σ^2

- $\frac{\partial l}{\partial\sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(X_i-\mu_0)^2 = 0 \Rightarrow \sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu_0)^2}$ is the MLE

  Let $Y = X-\mu_0 \sim N(0,\sigma^2)$ $\quad T = \sum_{i=1}^{n}Y_i^2$ $\quad \frac{T}{\sigma^2} \sim \chi^2_n$

  $w = \frac{T}{\sigma^2}$, $E\left(\frac{T}{\sigma^2}\right)^{\frac{1}{2}} = \int_0^{\infty}\frac{w^{\frac{n}{2}-1+\frac{1}{2}}e^{-\frac{w}{2}}}{2^{n/2}\Gamma(\frac{n}{2})}dw = \int_0^{\infty}\frac{w^{\frac{n+1}{2}-1}e^{-w}}{2^{\frac{n+1}{2}}\Gamma(\frac{n+1}{2})}dw \cdot \frac{2^{\frac{1}{2}}\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})} = \sqrt{2}\,\frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})}$

  $\therefore$ $E\,T^{\frac{1}{2}}\cdot\frac{\Gamma(\frac{n}{2})}{\sqrt{2}\,\Gamma(\frac{n+1}{2})} = \sigma$

  $\therefore$ $\left[\sum_{i=1}^{n}(X_i-\mu_0)^2\right]^{\frac{1}{2}}\cdot\frac{\Gamma(\frac{n}{2})}{\sqrt{2}\,\Gamma(\frac{n+1}{2})}$ is the UMVUE of σ

6. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Derive the UMVUE of (1) $\mu + \sigma^2$, and (2) μ^2/σ^2.

since the joint pdf $f(\vec{X},\theta) = (2\pi\sigma^2)^{-\frac{n}{2}}\exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2\right\}$ belongs to exponential family

$T(X) = \sum_{i=1}^{n}X_i$ $\quad G(X) = \sum_{i=1}^{n}(X_i-\bar{X})^2$ are sufficient & complete statistics of μ & σ^2

– $E\left[\frac{1}{n}T(X) + \frac{1}{n-1}G(X)\right] = \frac{1}{n}\cdot n\mu + \frac{1}{n-1}(n-1)\sigma^2 = \mu+\sigma^2$

$\therefore \frac{1}{n}\sum_{i=1}^{n} X_i + \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \frac{1}{n}\sum_{i=1}^{n}X_i)^2$ is the UMVUE of $\mu+\sigma^2$

– $ET^2(X) = nEX^2 + n(n-1)\mu^2 = n(\sigma^2+\mu^2) + n(n-1)\mu^2 = n^2\mu^2 + n\sigma^2$

$\frac{G(X)}{\sigma^2} \sim \chi^2_{n-1}$ let $W = \frac{G(X)}{\sigma^2} \sim \chi^2_{n-1}$

$E\frac{1}{G(X)} = \frac{1}{\sigma^2}\int_0^{+\infty} \frac{w^{\frac{n-1}{2}-2}e^{-\frac{w}{2}}}{2^{\frac{n-1}{2}}\Gamma(\frac{n-1}{2})}dx = \frac{1}{\sigma^2}\int_0^{+\infty}\frac{w^{\frac{n-3}{2}-1}e^{-\frac{w}{2}}}{2^{\frac{n-3}{2}}\Gamma(\frac{n-3}{2})}dx \cdot \frac{\Gamma(\frac{n-3}{2})}{2\Gamma(\frac{n-1}{2})} = \frac{1}{(n-3)\sigma^2}$

$E\frac{T^2(X)}{G(X)} = \frac{n^2}{n-3}\frac{\mu^2}{\sigma^2} + \frac{n}{n-3}$

so the UMVUE of $\frac{\mu^2}{\sigma^2}$ is $\frac{(n-3)T^2(X)}{n^2 G(X)} - \frac{1}{n}$

7. Let $X_1, \cdots, X_m$ i.i.d. $\sim N(\mu, \sigma^2)$, $Y_1, \cdots, Y_n$ i.i.d. $\sim N(2\mu, \sigma^2)$, and suppose that X_i's and Y_j's are independent, derive the UMVUE of μ and σ^2.

the joint pdf is $f(\vec{x}, \vec{y}; \mu, \sigma^2) = (2\pi\sigma^2)^{-\frac{m+n}{2}} \exp\left\{\frac{1}{2\sigma^2}\left[\sum_{i=1}^{m}(x_i-\mu)^2 + \sum_{i=1}^{n}(y_i-2\mu)^2\right]\right\}$

belongs to exponential family $f = (2\pi\sigma^2)^{-\frac{m+n}{2}} \exp\left\{\frac{1}{2\sigma^2}\left[n^2(4n+m) + \sum_{i=1}^{m}x_i^2 + \sum_{i=1}^{n}y_i^2 - 2n\sum_{i=1}^{m}x_i - 4n\sum_{i=1}^{n}y_i\right]\right\}$

$n \in R$ $\sigma^2 > 0$ has interior points of R^2

so, $T_1(X) = \sum_{i=1}^{m}X_i^2 + \sum_{j=1}^{n}Y_j^2$ $T_2(X) = \sum_{i=1}^{m}X_i + 2\sum_{j=1}^{n}Y_j$ are sufficient & complete statistic

$ET_2(X) = m\mu + 4n\mu$ thus, the UMVUE of μ is $\frac{T_2(X)}{m+4n}$

$ET_1(X) = m(\sigma^2+\mu^2) + n(\sigma^2+4\mu^2) = (m+n)\sigma^2 + (m+4n)\mu^2$

$ET_2^2(X) = (m+4n)^2\mu^2 + (m+4n)\sigma^2$

thus, the UMVUE of σ^2 is $\frac{1}{m+n-1}\left(T_1 - \frac{T_2^2}{m+4n}\right)$

8. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$, where $-\infty < \mu < \infty$ and $\sigma^2 > 0$ are unknown.

   – Derive the moment estimator and MLE of σ^2;
   – Derive the C-R lower bound for the variance of the unbiased estimator of σ^2;
   – Derive the UMVUE of σ^2.

– $S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$ $\bar{X} = \sum_{i=1}^{n}X_i$ $ES^2 = \sigma^2$

$\therefore$ the moment estimator of σ^2 is S^2

Likelihood function $f(\mu, \sigma^2, \vec{x}) = (2\pi\sigma^2)^{-\frac{n}{2}}\exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right\}$

$l = -\frac{n}{2}\ln 2\pi\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$

$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\bar{x})^2 = 0$

since it is in exponential family, the mle of σ^2 is $\frac{1}{n}\sum(x_i-\bar{x})^2 = S_n^2$

- $I(\sigma^2) = E\left[\frac{\partial \ln f(x_i,\mu,\sigma^2)}{\partial \sigma^2}\right]^2 = \frac{1}{4\sigma^8}E\left[(x-\mu)^2-\sigma^2\right]^2 = \frac{1}{4\sigma^8}\mathrm{var}(x-\mu)^2 = \frac{1}{2\sigma^4}$

  $\gamma(\sigma^2) = \sigma^2$

  C-R lower bound is $\frac{2}{n}\sigma^4$

- $T(x) = \sum_{i=1}^{n}(x_i-\bar{x})^2$ is sufficient & complete statistic of σ^2

  $\frac{T(x)}{\sigma^2} \sim \chi^2_{n-1}$ $\quad E\frac{1}{n-1}T(x) = \sigma^2$

  thus, the UMVUE of σ^2 is $\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$

9. Let $(X_1, Y_1), \cdots, (X_n, Y_n)$ i.i.d. drawn from a Bivariate Normal distribution $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.
   - Derive the MLEs of $(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$;
   - Calculate the observed values of the above MLEs using the data in Homework 2.

- $f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\left\{-\frac{\frac{(x-\mu_1)^2}{\sigma_1^2}+\frac{(y-\mu_2)^2}{\sigma_2^2}-\frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}}{2(1-\rho^2)}\right\}$

  likelihood function $L(\theta; \vec{x}, \vec{y}) = (2\pi\sigma_1\sigma_2\sqrt{1-\rho^2})^{-n}\exp\left\{-\frac{\frac{1}{\sigma_1^2}\sum(x_i-\mu_1)^2+\frac{1}{\sigma_2^2}\sum(y_i-\mu_2)^2-\frac{2\rho}{\sigma_1\sigma_2}\sum(x_i-\mu_1)(y_i-\mu_2)}{2(1-\rho^2)}\right\}$

  $l(\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho) = -n\ln(2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}) - \frac{\frac{1}{\sigma_1^2}\sum(x_i-\mu_1)^2+\frac{1}{\sigma_2^2}\sum(y_i-\mu_2)^2-\frac{2\rho}{\sigma_1\sigma_2}\sum(x_i-\mu_1)(y_i-\mu_2)}{2(1-\rho^2)}$

  belongs to exponential family

  $\frac{\partial l}{\partial \mu_1} = -\frac{1}{2(1-\rho^2)\sigma_1^2}\left(-2\sum_{i=1}^{n}x_i + 2\mu_1 n\right) + \frac{\rho}{(1-\rho^2)\sigma_1\sigma_2}\sum_{i=1}^{n}(\mu_2-y_i) = 0$

  $\frac{\partial l}{\partial \mu_2} = -\frac{1}{2(1-\rho^2)\sigma_2^2}\left(-2\sum_{i=1}^{n}y_i + 2\mu_2 n\right) + \frac{\rho}{(1-\rho^2)\sigma_1\sigma_2}\sum_{i=1}^{n}(\mu_2-y_i) = 0$

  $\frac{\partial l}{\partial \sigma_1^2} = -\frac{n}{2}\cdot\frac{1}{\sigma_1^2} + \frac{1}{\sigma_1^4}\sum_{i=1}^{n}(x_i-\mu_1)^2 - \frac{\rho}{2(1-\rho^2)\sigma_1\sigma_2}\sum_{i=1}^{n}(x_i-\mu_1)(y_i-\mu_2) = 0$

  $\frac{\partial l}{\partial \sigma_2^2} = -\frac{n}{2}\cdot\frac{1}{\sigma_2^2} + \frac{1}{\sigma_2^4}\sum_{i=1}^{n}(y_i-\mu_2)^2 - \frac{\rho}{2(1-\rho^2)\sigma_1\sigma_2}\sum_{i=1}^{n}(x_i-\mu_1)(y_i-\mu_2) = 0$

  $\frac{\partial l}{\partial \rho} = \frac{n\rho}{1-\rho^2} - \frac{\rho}{(1-\rho^2)^2}\left[\frac{1}{\sigma_1^2}\sum(x_i-\mu_1)^2+\frac{1}{\sigma_2^2}\sum(y_i-\mu_2)^2\right] + \frac{1+\rho^2}{(1-\rho^2)^2}\cdot\frac{1}{\sigma_1\sigma_2}\sum_{i=1}^{n}(x_i-\mu_1)(y_i-\mu_2) = 0$

  $\Rightarrow \hat{\rho} = \frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-\bar{x})(y_i-\bar{y})}{\hat{\sigma}_1\hat{\sigma}_2}$ $\quad \hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n}x_i$ $\quad \hat{\mu}_2 = \frac{1}{n}\sum_{i=1}^{n}y_i$

  $\hat{\sigma}_1^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$ $\quad \hat{\sigma}_2^2 = \frac{1}{n}\sum(y_i-\bar{y})$

- calculate results:

  $\hat{\mu}_1 = 68.178$ $\quad \hat{\mu}_2 = 12.345$ $\quad \hat{\sigma}_1^2 = 4.15$ $\quad \hat{\sigma}_2^2 = 135$ $\quad \hat{\rho} = 0.33$

HW 4

1. The data set "data.csv" is a subset of the Ames House data set. The first row of this data set is the name of the variables, where "SalePrice" is the dependent variable (column 1), "Lot_Area" and "Central_Air" are two independent variable (predictors; column 2 and column 6, respectively). There are $n = 1598$ observations in total (rows 2–1599). Suppose that we use a linear regression model to approximate the relationship between SalePrice, Lot_Area, and Central_Air:

$$\text{SalePrice} = \beta_0 + \beta_1\ \text{Lot_Area} + \beta_2\ \text{Central_Air} + \epsilon.$$

Suppose that the noise ϵ_i i.i.d. $\sim N(0, \sigma^2)$. Calculate the MLEs of $\beta_0,\ \beta_1,\ \beta_2$, and σ^2.

since central-Air is discrete, if Y, map it as 1; N, map it as 0

= LINEST(A2:A1599, J2:K1599, TRUE, TRUE)

77888.13	4.422360714	63725.92
6777.696	0.318866984	7043.772
0.179674	72027.55575	#N/A
174.674	1595	#N/A
1.81E+12	8.27481E+12	#N/A

↓ SSresid

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)^2}$$

$$\hat{\beta}_2 = \frac{\sum_{i=1}^{n}(x_{2i} - \bar{x}_2)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{2i} - \bar{x}_2)^2}$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$\hat{\beta}_0 = 63725.92$ $\hat{\beta}_1 = 77888.13$ $\hat{\beta}_2 = 4.42236$ $\hat{\sigma}^2 = 5.1782 \times 10^9$

2. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample of size $n = 15$ from a normal distribution $N(\mu, 0.5^2)$, and the observed values are:

$$2.5, 2.7, 3.0, 2.8, 3.1, 2.7, 2.3, 2.8, 2.4, 2.8, 2.6, 2.6, 3.1, 3.0, 2.6.$$

Construct a 95% confidence interval for μ.

as σ^2 is known, $\hat{m} = \frac{1}{15}\sum_{i=1}^{15} x_i = 2.74$ $\quad \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

let z be $z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0,1)$

$$P\left(\left|\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}\right| \le z_{\alpha/2}\right) = 1 - \alpha \quad \Rightarrow \quad P_\mu\left(\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \le \mu \le \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) = 1 - \alpha$$

thus, $\left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2},\ \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right]$ is $1-\alpha$ confidence interval

$n = 15$, the interval is $[2.487, 2.993]$

3. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Normal distribution $N(\mu, \sigma^2)$, where μ and $\sigma > 0$ are unknown. To make $\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right]^{1/2}/4$ be a 95% upper confidence limit of σ, how large sample size n should be?

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ $\quad P\left(\frac{1}{4}\sqrt{(n-1)S^2} < \sigma\right) \ge 1 - \alpha$

$\Rightarrow P\left(\frac{(n-1)S^2}{\sigma^2} \le 16\right) \ge 0.95$ $\quad \therefore P\left(\chi^2_{n-1} \le 16\right) \ge 0.95 \quad \Rightarrow n \ge 27$

4. Suppose that $X_1, \cdots, X_m$ i.i.d. $\sim N(a, \sigma_1^2)$, $Y_1, \cdots, Y_n$ i.i.d. $\sim N(a, \sigma_2^2)$, where σ_1^2, σ_2^2 are known, a i unknown. Suppose that X_i's and Y_j's are independent.

   - Derive the UMVUE of a;
   - Construct a confidence interval for a with confidence level $1-\alpha$.

- Joint p.d.f $f(\vec{X}, \vec{y}, a, \sigma_1^2, \sigma_2^2) = (2\pi\sigma_1^2)^{-\frac{m}{2}} \exp\{-\frac{1}{2\sigma_1^2}\sum_{i=1}^{m}(x_i-a)^2\} (2\pi\sigma_2^2)^{-\frac{n}{2}} \exp\{-\frac{1}{2\sigma_2^2}\sum_{i=1}^{n}(y_i-a)^2\}$

$= (2\pi\sigma_1^2)^{-\frac{m}{2}} (2\pi\sigma_2^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma_1^2}\left[\sum x_i^2 - 2a\sum x_i + ma^2\right] - \frac{1}{2\sigma_2^2}\left[\sum y_i^2 - 2a\sum y_i + na^2\right]\right\}$

$= (2\pi)^{-\frac{m+n}{2}} \sigma_1^{-m}\sigma_2^{-n} e^{-a^2(\frac{m}{2\sigma_1^2}+\frac{n}{2\sigma_2^2})} \exp\left\{a\left[\frac{1}{\sigma_1^2}\sum x_i + \frac{1}{\sigma_2^2}\sum y_i\right] - \frac{1}{2\sigma_1^2}\sum x_i^2 - \frac{1}{2\sigma_2^2}\sum y_i^2\right\}$

natural parameter $\theta^* \{a \mid a \in (-\infty, \infty)\}$ has interior points

thus, $T = \frac{1}{\sigma_1^2}\sum_{i=1}^{m} x_i + \frac{1}{\sigma_2^2}\sum_{i=1}^{n} y_i$ is complete & sufficient statistic of a

$ET = \frac{1}{\sigma_1^2} ma + \frac{1}{\sigma_2^2} na$ $\therefore$ the UMVUE of a is $\dfrac{\frac{1}{\sigma_1^2}\sum_{i=1}^{m} x_i + \frac{1}{\sigma_2^2}\sum_{i=1}^{n} y_i}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2}}$

- Let G be the UMVUE of a, $G \sim N\left(a, \dfrac{1}{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2}}\right)$

$P\left(\left|(G-a)\sqrt{\frac{m}{\sigma_1^2} + \frac{n}{\sigma_2^2}}\right| < N_{\frac{\alpha}{2}}\right) \le 1-\alpha$

thus, $P\left(G - \dfrac{N_{\frac{\alpha}{2}}}{\sqrt{\frac{m}{\sigma_1^2}+\frac{n}{\sigma_2^2}}} < a < G + \dfrac{N_{\frac{\alpha}{2}}}{\sqrt{\frac{m}{\sigma_1^2}+\frac{n}{\sigma_2^2}}}\right) \le 1-\alpha$

so the confidence interval is $\left[G - \dfrac{N_{\frac{\alpha}{2}}}{\sqrt{\frac{m}{\sigma_1^2}+\frac{n}{\sigma_2^2}}},\ G + \dfrac{N_{\frac{\alpha}{2}}}{\sqrt{\frac{m}{\sigma_1^2}+\frac{n}{\sigma_2^2}}}\right]$

5. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from the Uniform distribution $U(\theta-1/2, \theta+1/2)$, construct a confidence interval for θ with confidence level $1-\alpha$.

Joint p.d.f $f(\vec{X}, \theta) = 1 \cdot \prod_{i=1}^{n} I_{\theta-\frac{1}{2} < x_i < \theta+\frac{1}{2}} = I_{\theta-\frac{1}{2} \le X_{(1)} \le X_{(n)} \le \theta+\frac{1}{2}}$

thus, $X_{(1)}, X_{(n)}$ is sufficient statistic of θ

$X_{(1)}$: $f(x) = n(\theta+\frac{1}{2}-x)^{n-1} I_{\theta-\frac{1}{2} < x < \theta+\frac{1}{2}}$ $\quad X_{(n)}$: $f(x) = n(x-\theta+\frac{1}{2})^{n-1} I_{\theta-\frac{1}{2} < x < \theta+\frac{1}{2}}$

$EX_{(1)} = \frac{1}{2} + \theta - \frac{n}{1+n}$ $\quad EX_{(n)} = -\frac{1}{2} + \theta + \frac{n}{1+n}$ $\quad \therefore E\left[X_{(1)} + X_{(n)}\right] = 2\theta$

Let $Z = \dfrac{X_{(1)} + X_{(n)}}{2} - \theta$ Let $u = \dfrac{X_{(1)} - \theta}{2}$, $v = \dfrac{X_{(n)} - \theta}{2}$

$P(u, v) = n(n-1)\cdot 2^n (v-u)^{n-2} I_{-\frac{1}{2} \le u \le v \le \frac{1}{2}}$

$f(z) = \int_{-\infty}^{+\infty} P(u, z-u)du = \begin{cases} \int_{-\frac{1}{2}}^{\frac{1}{2}} 2^n n(n-1)(z-2u)^{n-2} du = n(1+2z)^{n-1}, & -\frac{1}{2} \le z \le 0 \\ n(1-2z)^{n-1}, & \text{or } z \le \frac{1}{2} \end{cases}$

$\therefore P\left(a \le \dfrac{X_{(1)} + X_{(n)}}{2} - \theta \le b\right) = \int_a^0 n(1+2z)^{n-1} dz + \int_0^b n(1-2z)^{n-1} dz$

$= 1 - \dfrac{(1+2a)^n + (1-2b)^n}{2} = 1-\alpha$

$\Rightarrow (1+2a)^n + (1-2b)^n = 2\alpha$

$P(\bar{z}-b \le \theta \le \bar{z}-a) = 1-\alpha$

interval length $L(a,b) = b-a$

using Lagrange method $L = b-a+\lambda\left((1+2a)^n + (1-2b)^n - 2\alpha\right)$

$\frac{\partial L}{\partial a} = -1 + 2n\lambda(1+2a)^{n-1}$ $\frac{\partial L}{\partial b} = 1 - 2n\lambda(1-2b)^{n-1}$ $\frac{\partial L}{\partial \lambda} = 2\alpha$

$\Rightarrow a = \frac{\sqrt[n]{\alpha}-1}{2}$ $b = \frac{1-\sqrt[n]{\alpha}}{2}$

thus, the confidence interval is $\left[\frac{X_{(1)}+X_{(n)}}{2} - \frac{1-\sqrt[n]{\alpha}}{2}, \frac{X_{(1)}+X_{(n)}}{2} - \frac{\sqrt[n]{\alpha}-1}{2}\right]$

6. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from the population with p.d.f. $f(x,\theta) = e^{-(x-\theta)},\ x > \theta$.
   - Show that the distribution of $X_{(1)} - \theta$ does not depend on θ;
   - Construct a confidence interval for θ with confidence level $1-\alpha$.

- $X_{(1)}$ p.d.f $f(x) = n\,e^{-(x-\theta)}(e^{-x+\theta})^{n-1} = n\,e^{n(\theta-x)}\,I_{x>\theta}$

  let $Y = X_{(1)} - \theta$ thus $f(y) = n\,e^{-ny}\,I_{y>0}$ not depend on θ

- $P(a \le X_{(1)} - \theta \le b) = \int_a^b ne^{-ny}\,dy = e^{-na} - e^{-nb} = 1-\alpha$

  to minimize $b-a$, $a = 0$ $b = \frac{-\ln\alpha}{n}$

  $\therefore$ confidence interval for θ is $\left[X_{(1)} + \frac{\ln\alpha}{n}, X_{(1)}\right]$

HW5

1. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from a distribution with unknown (finite) mean μ and known (finite) variance σ^2, and suppose that n is large. Then:

   (i) Construct a confidence interval for μ with confidence level approximately $1 - \alpha$.

   (ii) Provide the form of the interval in part (i) for $n = 100$, $\sigma = 1$, and $\alpha = 0.05$.

   (iii) Refer to part (i) and suppose that $\sigma = 1$ and $\alpha = 0.05$. Then determine the sample size n, so that the length of the confidence interval is less than 0.1.

   (iv) Show that the length of the interval in part (i) tends to 0 in probability as $n \to \infty$.

(i) $\bar{X}$ is a good point estimator of μ

$Z = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$ not depend on μ

$P\left(\left| \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \right| \le z_{\alpha/2} \right) = 1-\alpha$ thus $P_\mu\left(\bar{X} - \frac{\sigma z_{\alpha/2}}{\sqrt{n}} \le \mu \le \bar{X} + \frac{\sigma z_{\alpha/2}}{\sqrt{n}} \right) = 1-\alpha$

$\therefore \left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \right]$ is a confidence interval for μ

(ii) $\left[\bar{X} - 0.196, \bar{X} + 0.196 \right]$

(iii) $L = \frac{2\sigma}{\sqrt{n}} z_{\alpha/2} < 0.1 \Rightarrow n > 153$

(iv) $L = \frac{2\sigma}{\sqrt{n}} z_{\alpha/2}$ as $\sigma, z_{\alpha/2}$ are bounded, as $n \to \infty$ $L \to 0$

2. Consider the data set "data.csv" in Homework 4. There are 1475 houses which have central air-conditioner (Central_Air = Y) and 123 houses which don't have air-conditioner (Central_Air = N). Compare the mean SalePrice and the variance of SalePrice of the houses with central air-conditioner (Central_Air = Y) and those without central air-conditioner (Central_Air = N). More specifically, let $X_1, \cdots, X_{1475}$ denote the SalePrice of the houses with central air-conditioner and suppose that they are i.i.d. r.v.'s. Let $Y_1, \cdots, Y_{123}$ denote the SalePrice of the houses without central air-conditioner and suppose that they are i.i.d. r.v.'s too. Suppose that X_i's and Y_j's are independent.

   (i) Suppose that X_i *i.i.d.* $\sim N(\mu_1, \sigma^2)$, Y_j *i.i.d.* $\sim N(\mu_2, \sigma^2)$, $\sigma = 79400$. Construct a 95% confidence interval for $\mu_1 - \mu_2$.

   (ii) Suppose that X_i *i.i.d.* $\sim N(\mu_1, \sigma^2)$, Y_j *i.i.d.* $\sim N(\mu_2, \sigma^2)$, σ is unknown. Construct a 95% confidence interval for $\mu_1 - \mu_2$.

   (iii) Suppose that X_i *i.i.d.* $\sim N(\mu_1, \sigma_1^2)$, Y_j *i.i.d.* $\sim N(\mu_2, \sigma_2^2)$, $\sigma_1^2 \neq \sigma_2^2$ are unknown. Construct a 95% confidence interval for σ_1/σ_2.

   (iv) Refer to part (iii), construct approximately 95% confidence intervals for $\mu_1 - \mu_2$ by using central limit theory.

   (v) Refer to part (iii), construct 95% confidence region for (μ_1, μ_2).

   (vi) Plot the histogram of the SalePrice and its Gaussian Kernel density estimate (using R function "hist" and "density").

(i) $\bar{X} - \bar{Y}$ is a good point estimator for $\mu_1 - \mu_2$

$\bar{X} \sim N(\mu_1, \frac{\sigma^2}{m})$ $\bar{Y} \sim N(\mu_2, \frac{\sigma^2}{n})$ $Z = \frac{\bar{X} - \mu_1 - \bar{Y} + \mu_2}{\sigma\sqrt{\frac{1}{m}+\frac{1}{n}}} \sim N(0,1)$

$P(|Z| \le z_{\alpha/2}) = 1-\alpha$ $\therefore P\left(\bar{X} - \bar{Y} - \sigma\sqrt{\frac{1}{m}+\frac{1}{n}} \cdot z_{\alpha/2} \le \mu_1 - \mu_2 \le \bar{X} - \bar{Y} - \sigma\sqrt{\frac{1}{m}+\frac{1}{n}} \cdot z_{\alpha/2} \right) = 1-\alpha$

$\bar{X} = 180240.3$ $\bar{Y} = 101545.6$ $m = 1475$ $n = 123$

thus $[70095.4, 99306.3]$ is a 95% confidence interval for $\mu_1 - \mu_2$

(ii) $S_X^2 = \frac{1}{m-1}\sum_{i=1}^{m}(X-\bar{X})^2$ $\quad S_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X-\bar{X})^2$

$\frac{(m-1)S_X^2}{\sigma^2} + \frac{(n-1)S_Y^2}{\sigma^2} \sim \chi^2_{m+n-2}$ $\quad \therefore \frac{\bar{X}-\bar{Y}-\mu_1+\mu_2}{\sqrt{\frac{(m-1)S_X^2+(n-1)S_Y^2}{m+n-2}\left(\frac{1}{m}+\frac{1}{n}\right)}} \sim t_{m+n-2}$

$P(|T| \le t_{\alpha/2}) = 1-\alpha$ $\quad$ Let $S_W^2 = \frac{1}{m+n-2}\left[(m-1)S_X^2 + (n-1)S_Y^2\right]$

$\therefore \left[\bar{X}-\bar{Y} - S_W\, t_{m+n-2}(\frac{\alpha}{2})\sqrt{\frac{1}{m}+\frac{1}{n}},\ \bar{X}-\bar{Y} + S_W\, t_{m+n-2}(\frac{\alpha}{2})\sqrt{\frac{1}{m}+\frac{1}{n}}\right]$

$S_W \doteq 76223.1$

$\therefore [70669, 98732]$ is a 95% confidence interval for $\mu_1-\mu_2$

(iii) $F = \frac{S_Y^2}{\sigma_2^2} \Big/ \frac{S_X^2}{\sigma_1^2} \sim F_{n-1,m-1}$

$P(a \le F \le b) = 1-\alpha$ thus $P\left(\sqrt{a}\frac{S_X}{S_Y} \le \frac{\sigma_1}{\sigma_2} \le \sqrt{b}\frac{S_X}{S_Y}\right) = 1-\alpha$

$a = F_{n-1,m-1}(1-\frac{\alpha}{2})$ $\quad b = F_{n-1,m-1}(\frac{\alpha}{2})$

$\therefore [2.178, 2.830]$ is a 95% confidence interval for $\frac{\sigma_1}{\sigma_2}$

(iv) using CLT $\quad \frac{\bar{Y}-\bar{X}-(\mu_1-\mu_2)}{\sqrt{S_X^2/m + S_Y^2/n}} \sim N(0,1)$

$$\left[\bar{X}-\bar{Y} - z_{\alpha/2}\cdot\sqrt{\frac{S_X^2}{m}+\frac{S_Y^2}{n}},\ \bar{X}-\bar{Y} + z_{\alpha/2}\cdot\sqrt{\frac{S_X^2}{m}+\frac{S_Y^2}{n}}\right]$$

thus $[77766.6, 91634.8]$ is a 95% confidence interval for $\mu_1-\mu_2$

(v) using Bonferroni correction

$\frac{\sqrt{m}(\bar{X}-\mu_1)}{S_X} \sim t_{m-1}$ $\quad \frac{\sqrt{n}(\bar{Y}-\mu_2)}{S_Y} \sim t_{n-1}$

(μ_1,μ_2) confidence region is $\left[\bar{X} - t_{m-1,\frac{\alpha}{4}}\frac{S_X}{\sqrt{m}}, \bar{X} + t_{m-1}(\frac{\alpha}{4})\frac{S_X}{\sqrt{m}}\right] \times \left[\bar{Y} - t_{m-1}(\frac{\alpha}{4})\frac{S_Y}{\sqrt{n}}, \bar{Y} + t_{m-1}\frac{S_Y}{\sqrt{n}}\right)$

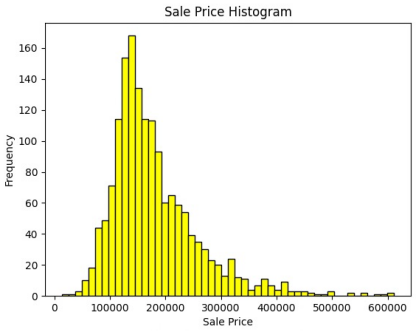
thus $[182223, 190270] \times [98722, 104369]$ is 95% confidence region

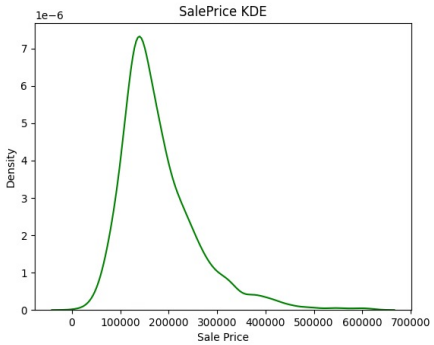
(vi) since X, Y independent, we need Šidák correction

$[181654, 190839] \times [98019, 105092]$

using python to paint, following the code

```python
import pandas as pd
import numpy as np
from sklearn.neighbors import KernelDensity
import matplotlib.pyplot as plt

csv_path = "./data.csv"
data = pd.read_csv(csv_path)

sale_price = data["SalePrice"]

# histogram
plt.hist(sale_price, bins=50, color="yellow", edgecolor="black")
plt.xlabel("Sale Price")
plt.ylabel("Frequency")
plt.title("Sale Price Histogram")
plt.show()

# kernel density estimation

kde = KernelDensity(kernel="gaussian", bandwidth=1000).fit(
    sale_price.values.reshape(-1, 1)
)

x = np.linspace(0, 620000, 10000)
log_dens = kde.score_samples(x.reshape(-1, 1))
plt.plot(x, np.exp(log_dens), color="green")
plt.xlabel("Sale Price")
plt.ylabel("Density")
plt.title("Sale Price KDE")
plt.show()
```





3. In the following examples, indicate which statements constitute a simple and which a composite hypothesis:

   (i) When tossing a coin, let X be the r.v. taking value 1 if the head appears and 0 if the tail appears. The statement is: The coin is biased.

   (ii) X is a r.v. whose expectation is equal to 5.

(i) composite (ii) simple

4. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from a binomial distribution $B(1, p)$ where p is unknown and $n = 20$. To test: $H_0 : p = 0.1 \leftrightarrow H_1 : p \neq 0.1$, we use the following test function:

$$\varphi(\boldsymbol{X}) = \begin{cases} 1, & \sum_{i=1}^{20} X_i \geq 6 \text{ or } \sum_{i=1}^{20} X_i \leq 2, \\ 0, & \text{otherwise}. \end{cases}$$

   (i) Calculate the power function of the test at $p = 0, 0.1, 0.2, \cdots, 0.9, 1$.

   (ii) Determine the level of significance α and the probability of type II error.

(i) $\pi_\varphi(p) = E_\theta \varphi(X) = 1 - \left[C_{20}^3 p^3 (1-p)^{17} + C_{20}^4 p^4 (1-p)^{16} + C_{20}^5 p^5 (1-p)^{15} \right]$

calculate by python. math

keep 10 decimal places

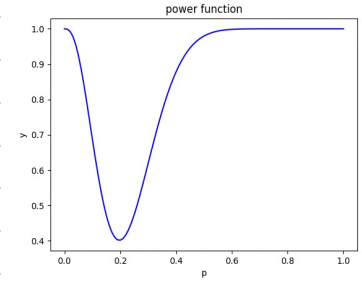
```
p =  0.0 power_function =  1.0000000000
p =  0.1 power_function =  0.6881799394
p =  0.2 power_function =  0.4018769335
p =  0.3 power_function =  0.6191123029
p =  0.4 power_function =  0.8780124993
p =  0.5 power_function =  0.9795064926
p =  0.6 power_function =  0.9983935166
p =  0.7 power_function =  0.9999570977
p =  0.8 power_function =  0.9999998198
p =  0.9 power_function =  1.0000000000
p =  1.0 power_function =  1.0000000000
```

(ii)

level of significance $\alpha = \pi_\varphi(0.1) = 0.6881$

type II error

$$\beta(p) = \begin{cases} 1 - \pi_\varphi(p), & p \neq 0.1 \\ 0, & p = 0.1 \end{cases}$$



5. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, 4)$ where μ is unknown. Let $\bar{X}$ be the sample mean. To test: $H_0 : \mu = \mu_0 \leftrightarrow H_1 : \mu \neq \mu_0$, the rejection region is

$$D = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : |\bar{X} - \mu_0| \geq c \right\}.$$

(i) Determine c such that the level of significance $\alpha = 0.05$.

(ii) Determine the power function of the test.

(iii) How does the power function change with sample size n?

(i) $Z = \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} \sim N(0,1) \mid H_0$

$\alpha = P(|\bar{X} - \mu_0| \geq c) = P\left(\left|\frac{\sqrt{n}(\bar{X}-\mu_0)}{2}\right| \geq \frac{c\sqrt{n}}{2}\right) = 0.05$

$\therefore \frac{c\sqrt{n}}{2} = z_{0.025} \Rightarrow c = \frac{2z_{0.025}}{\sqrt{n}}$

(ii) considering $Z = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$

power function $\pi_\varphi(\mu) = P(|\bar{X} - \mu_0| \geq c) = P\left(\frac{\sqrt{n}}{\sigma}|\bar{X} - \mu + \mu - \mu_0| \geq \frac{\sqrt{n}c}{\sigma}\right)$

$= P\left|\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \geq \frac{\sqrt{n}}{\sigma}(c + \mu_0 - \mu)\right) + P\left|\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \leq \frac{\sqrt{n}}{\sigma}(-c + \mu_0 - \mu)\right)$

$= 1 - \Phi\left(\frac{\sqrt{n}}{2}(c + \mu_0 - \mu)\right) + \Phi\left(\frac{\sqrt{n}}{2}(-c + \mu_0 - \mu)\right)$

(iii) As n increasing ② $\mu \neq \mu_0$ $\frac{2\pi}{\sigma\mu}$ $\Phi\left[\frac{\sqrt{n}}{2}(c + \mu_0 - \mu)\right] - \Phi\left[\frac{\sqrt{n}}{2}(-c + \mu_0 - \mu)\right]$ become less

① $\mu = \mu_0$ constant so power function is more kurtosis $\wedge \rightarrow \wedge$

6. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from uniform distribution $U(0, \theta)$, where θ is unknown. Let $X_{(n)} = \max\{X_1, \cdots, X_n\}$. For the hypothesis testing problem $H_0 : \theta \leq \theta_0, \leftrightarrow \ H_1 : \theta > \theta_0$, consider the following test function:

$$\varphi(\boldsymbol{x}) = \begin{cases} 1, & X_{(n)} \geq c, \\ 0, & \text{otherwise}. \end{cases} \tag{1}$$

(i) Calculate the power function of the test $\varphi(\boldsymbol{x})$, and show that it is an increasing function of θ.

(ii) If $\theta_0 = 1/2$, determine the critical value c such that the level of significance $\alpha = 0.05$.

(iii) Refer to (ii), determine the sample size n such that the power of the test at $\theta = 3/4$ is at least 0.98.

(iv) Refer to (ii), determine the sample size n such that the type II error of the test is no more than 0.02 at $\theta = 3/4$.

(i) the pdf of $X_{(n)}$ is $f(x) = n\frac{x^{n-1}}{\theta^n} I_{0<x<\theta}$

$E_\theta\varphi(x) = \begin{cases} \int_c^\theta n\frac{x^{n-1}}{\theta^n}dx = 1 - \left(\frac{c}{\theta}\right)^n, & 0<c<\theta \\ 1, & c<0 \\ 0, & c>\theta \end{cases}$ as θ increasing, power function increase

(ii) $\alpha = 1 - (2c)^n = 0.05 \Rightarrow c = \frac{\sqrt[n]{0.95}}{2}$

(iii) $1 - 0.95\left(\frac{3}{4}\right)^n = 0.98 \Rightarrow n = 10$

(iv) $1 - 1 + \left(\frac{c}{\theta}\right)^n = 0.02 \Rightarrow n = 10$

7. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$ where μ is unknown and σ is known.

   (i) For testing the hypothesis $H_0 : \mu \leq 0 \;\leftrightarrow\; H_1 : \mu > 0$, show that the sample size n can be determined to achieve a given level of significance α and given power $\pi(1)$.

   (ii) What is the numerical value of n for $\alpha = 0.05, \pi(1) = 0.9$ when $\sigma = 1$?

(i) given H_0 $\quad Z = \frac{\sqrt{n}\bar{X}}{\sigma} \sim N(0,1)$

rejection region $D = \{\vec{X} : \sqrt{n}\bar{X}/\sigma > c\}$ $\quad P\left(\frac{\sqrt{n}\bar{X}}{\sigma} > c\right) = \alpha$

power function $\pi(\mu) = P_\mu\left(\frac{\sqrt{n}\bar{X}}{\sigma} > c\right) = P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} > c - \frac{\sqrt{n}\mu}{\sigma}\right) = 1 - \Phi\left(c - \frac{\sqrt{n}\mu}{\sigma}\right)$

$\pi(\mu)$ is increasing as μ increases

$\alpha = \pi(0) = 1 - \Phi(c)$

given α & $\pi(1)$, we can determin n

(ii) $\begin{cases} \Phi(c) = 0.95 \\ 1 - \Phi(c - \sqrt{n}) = 0.9 \end{cases} \Rightarrow \begin{cases} c = 1.64 \\ \sqrt{n} - c = 1.29 \end{cases} \Rightarrow n = 9$

8. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Normal distribution $N(\mu, \sigma^2)$ where μ is unknown and σ^2 is known. For testing the hypothesis $H_0 : \mu \geq \mu_0 \;\longleftrightarrow\; H_1 : \mu < \mu_0$. Show that the rejection region with level of significance α is given by

$$D_3 = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} < -z_\alpha \right\}$$

the rejection region is $\{\vec{X} = (X_1, \cdots X_n) : Z = \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} < c\}$

$\sup\{P(Z < c \mid H_0)\} = \sup\{P(Z < c \mid \mu \geq \mu_0)\} = \alpha$

$P_\mu(Z < c) = P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} < c + \frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right) = 1 - \Phi\left(c + \frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right)$

increase as μ increasing

thus, $\alpha = P(Z < c \mid \mu = \mu_0) = P\left(\frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} < c\right) \Rightarrow c = -z_\alpha$

$\therefore D_3 = \left\{\vec{X} = (X_1 \cdots X_n) : Z = \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} < -z_\alpha\right\}$ is the rejection region with α

9. A manufacturer claims that packages of certain goods contain 18 ounces. In order to check his claim, 100 packages are chosen at random from a large lot and it is found that $\sum_{i=1}^{100} x_i = 1570$ and $\sum_{i=1}^{100} x_i^2 = 32000$. Assume that the observations are Normally distributed, and formulate the manufacturer's claim as a testing hypothesis problem. Carry out the test at level of significance $\alpha = 0.01$ and compute the p-value.

$H_0 : \mu = \mu_0 \;\leftrightarrow\; H_1 : \mu \neq \mu_0$

$T = \frac{\sqrt{n}(\bar{X}-\mu_0)}{S} \Big|_{\mu=\mu_0} \sim t_{n-1}$ $\quad$ rejection region $\{\vec{X} : |T| > t_{n-1, \frac{\alpha}{2}}\}$

where $\alpha = 0.01$ $\quad n = 100$ $\quad \mu_0 = 18$ $\quad t_{99, 0.005} = 2.626$

$\bar{X} = 15.7$ $\quad S^2 = 74.45$ $\quad$ thus $|T| = \left|\frac{10(15.7-18)}{8.617}\right| = 2.67 > 2.626$

so we reject H_0, $\quad p\text{-value} = P(|t_{99}| > 2.67 \mid \mu = 18) = 0.0089 < \alpha$ $\quad$ thus we reject H_0

HW 6

1. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$ where μ is unknown and σ is known.
   - For testing the hypothesis $H_0 : \mu \leq 0 \leftrightarrow H_1 : \mu > 0$, show that the sample size n can be determined to achieve a given level of significance α and given power $\pi(1)$.
   - What is the numerical value of n for $\alpha = 0.05,\ \pi(1) = 0.9$ when $\sigma = 1$?

(i) given H_0 $\quad Z = \frac{\sqrt{n}\bar{X}}{\sigma} \sim N(0,1)$

rejection region $D = \{\vec{X} : \sqrt{n}\bar{X}/\sigma > c\}$ $\quad P\left(\frac{\sqrt{n}\bar{X}}{\sigma} > c\right) = \alpha$

power function $\pi(\mu) = P_\mu\left(\frac{\sqrt{n}\bar{X}}{\sigma} > c\right) = P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} > c - \frac{\sqrt{n}\mu}{\sigma}\right) = 1 - \Phi\left(c - \frac{\sqrt{n}\mu}{\sigma}\right)$

$\pi(\mu)$ is increasing as μ increases

$\alpha = \pi(0) = 1 - \Phi(c)$

given α & $\pi(1)$, we can determin n

(ii) $\begin{cases} \Phi(c) = 0.95 \\ 1 - \Phi(c - \sqrt{n}) = 0.9 \end{cases} \Rightarrow \begin{cases} c = 1.64 \\ \sqrt{n} - c = 1.29 \end{cases} \Rightarrow n = 9$

2. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from binomial distribution $B(1, \theta)$.
   - Considering the test problem $H_0 : \theta \leq 0.01 \leftrightarrow H_1 : \theta > 0.01$, construct a test with level of significance $\alpha = 0.05$.
   - Determine the sample size n such that the type II error of the above test is no more than 0.01 at $\theta = 0.08$.

- $\bar{X}$ is a good estimator of θ $\quad T(X) = \sum_{i=1}^{n} X_i$

the rejection region can take the form $D = \{\vec{X} \mid T(X) > c\}$

$T(X) \sim B(n, \theta)$ $\quad H_0^*: \theta = 0.01$

$P(T(X) > c \mid H_0^*) = P_{0.01}(T(X) > c) = \alpha$

using a randomized test, $\varphi(X) = \begin{cases} 1, & T(X) > c \\ r, & T(X) = c \\ 0, & T(X) < c \end{cases}$

thus, $E_\theta \varphi(X) = P_{0.01}(T(X) > c) + r\, P_{0.01}(T(X) = c) = \alpha$

power function $\pi(\theta) = P_\theta(T(X) > c) + r\, P_\theta(T(X) = c)$ increasing of θ

thus $E_{0.01}\varphi(X) = \alpha$ $\quad P_\theta(T(X) = c) = C_n^c \theta_0^c (1-\theta_0)^{n-c}$

where c is the maximum integer when $P_\theta(T(X) > c) < \alpha$

and $r = \frac{\alpha - P_\theta(T(X) > c)}{P_\theta(T(X) = c)}$

$P_\theta(T(X) > c)$ is increasing as θ, thus $\alpha = \sup\{\pi(p), p \leq 0.01\} = \pi(0.01) \leq 0.05$

- type II error $\beta = 1 - \pi(0.08) < 0.01$

$\Rightarrow 1 - (1 - \sum_{k=0}^{c} \binom{n}{k} 0.08^k 0.92^{n-k} + r\binom{n}{c} 0.08^c 0.92^{n-c}) \le 0.01$

we can obtain n, using python enum, $n \ge 120$

3. The breaking powers of certain steel bars produced by processes A and B are independent r.v.'s distributed at Normal with possibly different means but the same variance. A random sample of size 25 is taken from bars produced by each one of the processes, and it is found that $\bar{x} = 60, s_x = 5, \bar{y} = 70, s_y = 6$. Test whether there is a difference between the two processes at the level of significance $\alpha = 0.05$.

$H_0: \mu_1 - \mu_2 = 0 \iff H_1: \mu_1 - \mu_2 \neq 0$

since $\bar{X} - \bar{Y}$ is a good estimator of $\mu_1 - \mu_2$, considering rejection region $D = \{(\vec{X}, \vec{Y}) \mid |\bar{X} - \bar{Y} - \mu_0| > A\}$

To determine A, under H_0, $T = \dfrac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}} \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}$

thus $P(|T| > t_{m+n-2}(\frac{\alpha}{2})) = 1 - \alpha$

rejection region $D = \{(\vec{X}, \vec{Y}) \mid |T| > t_{m+n-2}(\frac{\alpha}{2})\}$

$|T| = \left| \dfrac{70 - 60}{\sqrt{\frac{24 \times 25 + 24 \times 36}{48}} \cdot \sqrt{\frac{1}{25}}} \right| = 6.40 > 2.016$ so we reject H_0

there is a difference

4. An advertisement manager for a radio station claims that over $100p_0\%$ $(0 < p_0 < 1)$ of all young adults in the city listen to a weekend music program. To establish this conjecture, a random sample of size n is taken from among the target population and t of them listen to the weekend music program. Let $X_i = 1$ if the ith young adult listens to the program and $X_i = 0$ otherwise.
   - Decide a suitable probability model describing the number of young adults who listen to the weekend music program.
   - For $p_0 = 5\%$, $n = 100$, $t = 20$, check whether the claim made the manager is supported or not (at level of significance $\alpha = 0.01$).

- we consider Bionomial model, for each person $X \sim B(1, p)$

- $H_0: p \overset{\le}{\ge} p_0 \iff H_1: p \overset{>}{<} p_0$ $H_0 \iff H_1$

considering $T(X) = \sum_{i=1}^{n} X_i$ under H_0 $T(X) \sim (n, p_0)$

$D = \{\vec{X} \mid T(X) < c\}$ random test $\varphi(X) = \begin{cases} 1, & T(X) < c \\ r, & T(X) = c \\ 0, & T(X) > c \end{cases}$

$E_{p_0}\varphi(X) = P(T(X) < c) + r\,P(T(X) = c) = \alpha$

$\pi(p) = P(T(X) < c) + r\,P(T(X) = c)$ is increasing of p

thus $E_{p_0}\varphi(X) = \alpha$ $P(T(X) = c) = C_n^c p_0^c (1 - p_0)^{n-c}$

$n = 100$, $p_0 = 0.05$ $\alpha = 0.01$ $\Rightarrow c = 1$, $r = 0.869$

$T(X) = 20 > 1$ so we accept H_0

5. Two testers (A and B) analyzed the same product samples and obtained the results in Table 1. Test whether there is a significant difference (in means) of the analysis results of testers A and B. Take $\alpha = 0.05$.

Table 1: Testing results

tester	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8
X A	4.4	3.3	3.8	3.5	3.6	4.8	3.3	3.9
Y B	3.8	4.2	3.8	3.8	4.6	3.9	2.9	4.4

that's a paired comparison question, considering $Z_i = Y_i - X_i \sim N(\mu, \sigma^2)$

$H_0: \mu = 0 \leftrightarrow H_1: \mu \neq 0$ under H_0, $T = \frac{\sqrt{n}\,\bar{Z}}{S_Z} \sim t_{n-1}$

thus, the rejection region $D = \{ \vec{Z} \mid |T| < t_{n-1, \frac{\alpha}{2}} \}$

$\bar{Z} = 0.1$ $\quad S_Z^2 = 0.4857$ $\quad S_Z = 0.6969$ $\quad t_{7,0.025} = 2.3646$

$|T| = \left| \frac{\sqrt{8} \cdot 0.1}{0.6969} \right| = 0.406 < 2.3646$ thus, we accept H_0

there's no significant difference

6. The diameters of certain cylindrical items produced by a machine are r.v.'s distributed as $N(\mu, 0.04)$. A sample of size 16 is taken and it is found that $\bar{x} = 2.48$ inches.
   - If the desired value for μ is 2.5 inches, formulate the appropriate testing hypothesis problem and carry out the likelihood ratio test at level of significance $\alpha = 0.05$.
   - Determine the power of the test.

- $H_0: \mu = 2.5 \leftrightarrow H_1: \mu \neq 2.5$

likehood function $L(\vec{X}; \mu) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right\}$

$L_{MLE, H_0} = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right\}$

$L_{MLE, H} = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2 \right\}$

$-2\ln\lambda = \frac{1}{\sigma^2} \left[\sum (x_i - \mu_0)^2 - (x_i - \bar{x})^2 \right] = \left[\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} \right]^2$

$-2\ln\lambda > c^2$ if & only if $\left[\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} \right]^2 > c^2 \Leftrightarrow \left| \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} \right| > c$

under H_0, $\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \sim N(0,1)$ $\quad P_{\mu_0}\left(\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2} \right) = \alpha$

thus, the likelihood ratio test is $\varphi(\vec{X}) = \begin{cases} 1, & \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2} \\ 0, & \text{else} \end{cases}$

when $\mu_0 = 2.5$ $\bar{X} = 2.48$ $n = 16$ $\sigma^2 = 0.04$ $\alpha = 0.05$

$\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} = 0.4 < 1.96$ so we accept H_0 Let $Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0,1)$

- power function $\pi(\mu) = P\left(\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2} \right) = P\left(\left| \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} + \frac{\sqrt{n}(\mu - \mu_0)}{\sigma} \right| > z_{\alpha/2} \right)$

$= 1 - \Phi\left(z_{\alpha/2} - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma} \right) + \Phi\left(-z_{\alpha/2} - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma} \right) = 1 - \Phi(51.96 - 20\mu) + \Phi(48.04 - 20\mu)$

7. A coin, with probability θ of falling heads, is tossed independently 100 times and 60 heads are observed. At level of significance $\alpha = 0.01$:

   – Use the likelihood ratio test to test the hypothesis $H_0 : \theta = 1/2 \leftrightarrow H_1 : \theta \neq 1/2$.

   – Employ the appropriate approximation to determine the critical value.

— $L(\theta; \vec{X}) = P(X_1 = i_1, X_2 = i_2 \cdots X_n = i_n) = C_n^t \theta^t (1-\theta)^{n-t}$, $(t = \sum_{k=1}^{n} i_k)$ $(i = 1/0)$

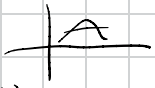
$L_{MLE,H_0} = C_n^t \theta_0^t (1-\theta_0)^{n-t}$ $\quad l = t\ln\theta + (n-t)\ln(1-\theta)$ $\quad \frac{\partial l}{\partial \theta} = 0 \Rightarrow \theta = \frac{t}{n}$

$L_{MLE,H} = (\frac{t}{n})^t (\frac{n-t}{n})^{n-t}$

$\lambda = (\frac{n\theta_0}{t})^t (\frac{n(1-\theta_0)}{n-t})^{n-t} > \lambda_0$

$\theta_0 = 1/2$ $\quad -\ln q = n\ln n + t\ln t + (n-t)\ln(n-t) = g(t)$

$g'(t) = \ln t - \ln(n-t) = 0 \Rightarrow t = \frac{n}{2}$



considering symmetry, $\varphi(T) = \begin{cases} 1, & T(X) < c_1 \text{ or } T(X) > c_2 \\ r_1, & T(X) = c_1 \\ r_2, & T(X) = c_2 \\ 0, & \text{else} \end{cases}$

$$\begin{cases} P(T(X) < c_1) + r_1 P(T(X) = c_1) = \frac{\alpha}{2} \\ P(T(X) > c_2) + r_2 P(T(X) = c_2) = \frac{\alpha}{2} \end{cases} \Rightarrow \begin{matrix} c_1 = 37 & r_1 = 0.623 \\ c_2 = 63 & r_2 = 0.623 \end{matrix}$$

when $\alpha = 0.01$ $\quad T(X) \mid H_0 \sim N(100, \frac{1}{2})$

$t = 60 \in (37, 63)$ $\quad$ so we accept H_0

(2) as showed above $\quad \varphi(T) = \begin{cases} 1, & T(X) < 37 \text{ or } T(X) > 63 \\ 0.623, & T(X) = 37 \\ 0.623, & T(X) = 63 \\ 0, & \text{else} \end{cases}$

CLT

8. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from uniform distribution $U(\theta, 1)$ where $\theta < 1$ is unknown.

   – At level of significance α, carry out the likelihood ratio test of the hypothesis

$$H_0 : \theta \geq \theta_0 \longleftrightarrow H_1 : \theta < \theta_0,$$

   where $\theta_0 < 1$ is given.

   – Determine the power function of the test.

— Likelihood function $L(\theta, \vec{X}) = (\frac{1}{1-\theta})^n I_{\theta < X_{(1)} < 1}$

$L_{MLE,H} = (\frac{1}{1-X_{(1)}})^n$ $\quad L_{MLE,H_0} = \begin{cases} (\frac{1}{1-X_{(1)}})^n, & \theta_0 \leq X_{(1)} \\ 0, & \text{else} \end{cases}$

$\lambda(\vec{X}) = \begin{cases} 1, & X_{(1)} \geq \theta_0 \\ 0, & \text{else} \end{cases}$ $\quad D = \{\vec{X} \mid X_{(1)} < c\}$

$\pi(\theta) = P(X_{(1)} < c) = \begin{cases} 1 - (\frac{1-c}{1-\theta})^n, & c \leq 1 \\ 1, & c > 1 \end{cases}$ decreasing by θ

$\alpha = \pi(\theta_0) \Rightarrow c = 1 - (1-\theta_0)(1-\alpha)^{\frac{1}{n}}$

— $\pi(\theta) = \begin{cases} 1 - (1-\alpha)(\frac{1-\theta_0}{1-\theta})^n, & \theta < c \\ 0, & \theta \geq c \end{cases}$

9. A medical researcher wishes to determine whether a pill has the undesirable side effect of reducing the blood pressure of the user. The study requires recording the initial blood pressure of n college-age women. After the use of the pill regularly for 6 months, their blood pressures are again recorded. With μ denoting the difference of blood pressure after the usage of the pill and before it, the claim is that $\mu < 0$.

   - Check this claim by testing the hypothesis $H_0 : \mu \geq 0 \;\leftrightarrow\; H_1 : \mu < 0$ at level of significance α, by using the likelihood ratio test.
   - Carry out the test if $n = 100$ and $\alpha = 0.05$.

— σ unknown $\quad L(\mu, \sigma^2; \vec{x}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right\}$

$H: \hat{\mu} = \bar{x} \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 \quad L(\hat{\Omega}) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\left\{-\frac{n}{2}\right\}$

$w = \{\mu, \sigma^2 \mid \mu \geq 0, \sigma^2 > 0\} \quad L(w) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\bar{x})^2 - \frac{n(\bar{x}-\mu)^2}{2\sigma^2}\right\}$

$g(\mu) = \exp\left\{-\frac{n(\bar{x}-\mu)^2}{2\sigma^2}\right\}$ $\quad \wedge_{\bar{x}}$

under H_0, when $\bar{x} \leq \mu_0$ $\quad L(\hat{w}) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\left\{-\frac{n}{2}\right\}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i-\mu_0)^2$

when $\bar{x} > \mu_0$ $\quad L(\hat{w}) = L(\hat{\Omega})$

thus, $\lambda = \begin{cases} 1, & \bar{x} > \mu_0 \\ \left[\dfrac{\sum(x_i-\bar{x})^2}{\sum(x_i-\mu_0)^2}\right]^{\frac{n}{2}}, & \bar{x} \leq \mu_0 \end{cases}$ Let $t = \dfrac{\sqrt{n}(\bar{x}-\mu_0)}{\sqrt{\frac{1}{n-1}\sum(x_i-\bar{x})^2}}$ $\sim t_{n-1} \mid \mu=\mu_0$

$\lambda = \left(1+\frac{t^2}{n-1}\right)^{-\frac{n}{2}}$ is an increasing function of t, $\lambda < \lambda' \Leftrightarrow t < t'$

$\therefore D = \{\vec{x} \mid T(x) < c\} \quad P_{\theta}(T(x) < c \mid \mu = \mu_0) = \alpha \Rightarrow c = -t_{n-1}(\alpha)$

$\pi(\mu) = P_{\mu}(T(x) < c) \leq P_{\mu_0}(T(x) < c) = \alpha$

thus $\varphi(x) = \begin{cases} 1, & T(x) < -t_{n-1}\alpha \\ 0, & \text{else} \end{cases}$, $\mu_0 = 0$

— $\varphi(x) = \begin{cases} 1, & \dfrac{10\bar{x}}{\sqrt{\frac{1}{99}\sum(x_i-\bar{x})^2}} < -1.66 \\ 0, & \text{else} \end{cases}$

10. Let $X_i, i = 1, \cdots, 9$ and $Y_j, j = 1, \cdots, 10$ be independent r.v.'s from the distributions $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively. Suppose that the observed values of the sample variance are $s_x^2 = 4$, $s_y^2 = 9$.

   - At level of significance $\alpha = 0.05$, test the hypothesis $H_0 : \sigma_1 = \sigma_2 \;\leftrightarrow\; H_1 : \sigma_1 \neq \sigma_2$ by using likelihood ratio test.
   - Find an expression for the power of the test for $\sigma_1 = 2$ and $\sigma_2 = 3$.

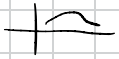
— $L(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2) = (2\pi\sigma_1^2)^{-\frac{m}{2}}(2\pi\sigma_2^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma_1^2}\sum_{i=1}^{m}(x_i-\mu_1)^2 - \frac{1}{2\sigma_2^2}\sum_{i=1}^{n}(y_i-\mu_2)^2\right\}$

$L(\hat{\Omega}) = (2\pi\hat{\sigma}_1^2)^{-\frac{m}{2}}(2\pi\hat{\sigma}_2^2)^{-\frac{n}{2}} \exp\left(-\frac{m+n}{2}\right)$

under H_0, $\hat{\mu}_1 = \bar{x}$, $\hat{\mu}_2 = \bar{y}$, $\hat{\sigma}^2 = \frac{1}{m+n}\left[\sum^{m}(x_i-\bar{x})^2 + \sum(y_i-\bar{y})^2\right]$

$L(\hat{w}) = (2\pi\hat{\sigma}^2)^{-\frac{m+n}{2}} \exp\left(-\frac{m+n}{2}\right)$ $\quad$ Let $a = \sum^{m}(x_i-\bar{x})^2$, $b = \sum^{n}(y_i-\bar{y})^2$

$\lambda(\vec{x}, \vec{y}) = \dfrac{(\frac{a}{m})^{-\frac{m}{2}} \cdot (\frac{b}{n})^{-\frac{n}{2}}}{(\frac{a+b}{m+n})^{-\frac{m+n}{2}}} = \dfrac{(m+n)^{\frac{m+n}{2}}}{m^{\frac{m}{2}} n^{\frac{n}{2}}} \cdot \dfrac{(\frac{m-1}{n-1}f)^{\frac{m}{2}}}{(1+\frac{m-1}{n-1}f)^{\frac{m+n}{2}}}$, $f = \dfrac{a}{m-1} \Big/ \dfrac{b}{n-1}$

$\lambda(T) < \lambda'$  $\Leftrightarrow T \le c_1$ or $T \ge c_2$ $T|_{H_0} \sim F_{m-1,n-1}$

thus, $c_1 = F_{m-1,n-1}(1-\frac{\alpha}{2})$ $c_2 = F_{m-1,n-1}(\frac{\alpha}{2})$

$$\varphi(X,Y) = \begin{cases} 1, & T \le c_1 \text{ or } T \ge c_2 \\ 0, & \text{else} \end{cases}$$

when $m=9$, $n=10$ $S_X^2 = 4$ $S_Y^2 = 9$ $\alpha = 0.05$

$T = \frac{S_X^2}{S_Y^2} = \frac{4}{9}$ $c_1 = 0.23$ $c_2 = 4.10$ so we accept H_0

$\pi(\theta) = P(T \le c_1 \text{ or } T \ge c_2)$ $T = \frac{\frac{1}{m-1}\sum(X_i-\bar{X})^2}{\frac{1}{n-1}\sum(Y_i-\bar{Y})^2} \cdot \frac{\sigma_2^2}{\sigma_1^2}$

$= 1 - P(c_1 \le \frac{\sigma_2^2}{\sigma_1^2} F_{m-1,n-1} \le c_2)$

when $\sigma_1 = 2$ $\sigma_2 = 3$ $\pi(\theta) = 0.184$

HW 7

1. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Normal distribution $N(\mu, \sigma^2)$, where μ is unknown and σ is known.
   - Derive the UMP test for testing the hypothesis $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu > \mu_0$ at level of significance α.
   - Carry out the testing hypothesis for $n = 100, \sigma^2 = 4, \mu_0 = 3, \bar{x} = 3.2, \alpha = 0.01$ and compute the power for $\mu = 3.5$.

– $H_0: \mu = \mu_0 \leftrightarrow H_1: \mu = \mu_1 (\mu_1 > \mu_0)$

$L_{H_0}(\mu; \vec{x}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\{-\frac{1}{2\sigma^2}\sum(x_i - \mu_0)^2\}$ $\quad L_{H_1}(\mu; \vec{x}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\{-\frac{1}{2\sigma^2}\sum(x_i - \mu_1)^2\}$

$\lambda = \frac{L_{H_0}(\mu; \vec{x})}{L_{H_1}(\mu; \vec{x})} = \exp\{-\frac{1}{2\sigma^2}\sum(\mu_0^2 - 2x_i(\mu_0 - \mu_1) + \mu_1^2)\} < c$

$\therefore D = \{\vec{x} : \bar{x} > c'\} \Leftrightarrow D = \{\vec{x} : \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > c''\}$ $\quad \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} \mid H_0 \sim N(0,1)$

$P(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > z_\alpha) = \alpha$ $\quad \varphi(x) = \begin{cases} 1, & \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > z_\alpha \\ 0, & \text{else} \end{cases}$ doesn't depend on μ_1

thus, the UMP is also $\varphi(x)$

– $\frac{\sqrt{100}(3.2 - 3)}{2} = 1 < 2.33 \ (z_{0.01})$ so we accept H_0

$\pi(\mu) = P(\frac{\sqrt{n}(\bar{x} - \mu + \mu - \mu_0)}{\sigma} > z_\alpha) = P(\frac{\sqrt{n}(\bar{x} - \mu)}{\sigma} > z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma})$

by numerical calculation, $\pi(3.5) = 1 - \Phi(2.326 - \frac{5}{2}) = 0.569$

2. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Normal distribution $N(\mu, \sigma^2)$, where μ is unknown and σ is known.
   - Derive the UMP test for testing the hypothesis $H_0 : \mu \leq \mu_0 \longleftrightarrow H_1 : \mu > \mu_0$ at level of significance α.
   - Carry out the test for $n = 100, \sigma^2 = 4, \mu_0 = 3, \bar{x} = 3.2, \alpha = 0.01$ and compute the power for $\mu = 3.5$.

– considering test $H_0'': \mu = \mu_0 \leftrightarrow H_1: \mu > \mu_0$

as illustrated above $\varphi(x) = \begin{cases} 1, & \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > z_\alpha \\ 0, & \text{else} \end{cases}$ is the UMP test

$H_0: \mu \leq \mu_0$ $\quad E_\theta \varphi(x) = P(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > z_\alpha) = 1 - \Phi(z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma})$

is an increasing function of μ

thus, $E_\theta \varphi(x) \leq E_\theta \varphi(\mu_0) = \alpha$, so it is also the UMP of test

– the same as above $\pi(3.5) = 0.569$

3. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from Gamma distribution Gamma(α_0, β) with α_0 known and β unknown.

   - Construct the MP test for testing the hypothesis $H_0 : \beta = \beta_1 \longleftrightarrow H_1 : \beta = \beta_2$ $(\beta_2 > \beta_1)$ at level of significance α.
   - Show that $X_1 + \cdots + X_n \sim \text{Gamma}(n\alpha_0, \beta)$.
   - Use the CLT to carry out the test when $n = 30, \alpha_0 = 10, \beta_1 = 2.5, \beta_2 = 3, \alpha = 0.05$ and compute the power.

- $L_{H_0}(\vec{X};\beta) = \prod_{i=1}^{n} \frac{\beta_1^{\alpha_0}}{\Gamma(\alpha_0)} x_i^{\alpha_0-1} e^{-\beta_1 x_i}, \ x>0 \qquad L_{H_1}(\vec{X};\beta) = \prod_{i=1}^{n} \frac{\beta_2^{\alpha_0}}{\Gamma(\alpha_0)} x_i^{\alpha_0-1} e^{-\beta_2 x_i}, \ x>0$

  $\lambda = \frac{L_{H_0}}{L_{H_1}} = \left(\frac{\beta_1}{\beta_2}\right)^{\alpha_0} e^{-(\beta_1-\beta_2)\sum x_i} < c \qquad \because \beta_2 > \beta_1$

  $D = \{ \vec{X} \mid \sum x_i < c \}$

  under H_0 $\quad \sum X_i \sim \text{Gamma}(n\alpha_0, \beta_1)$

  $P(\sum X_i < c) = \alpha \Rightarrow c = \text{Gamma}(n\alpha_0, \beta_1)(\alpha)$

  $\varphi(x) = \begin{cases} 1, & \sum x_i < \text{Gamma}(n\alpha_0, \beta_1)(1-\alpha) \\ 0, & \text{else} \end{cases}$ does not depend on β_1

  thus, it is also the UMP test

- mgf of Gamma(α, β) is $m(t) = (1-\beta t)^{-\alpha}$

  as for $\sum X_i$ $\quad m(t) = (1-\beta t)^{-n\alpha}$ the same as mgf of Gamma$(n\alpha, \beta)$

  thus $\sum x_i \sim \text{Gamma}(n\alpha_0, \beta)$

- by CLT, $\frac{\sqrt{n}(\bar{X} - \alpha\beta)}{\sqrt{\alpha\beta^2}} \xrightarrow{d} N(0,1)$ $\qquad \varphi(x) = \begin{cases} 1, & \frac{\sqrt{n}(\bar{X} - \alpha\beta)}{\sqrt{\alpha\beta^2}} < -z_\alpha \\ 0, & \text{else} \end{cases}$

  numerical calculate, under H_0 $\quad \varphi(x) = \begin{cases} 1, & \bar{X} < 3.62 \\ 0, & \text{else} \end{cases}$

  $\pi(\beta_2) = P\left(\frac{\sqrt{n}(\bar{X} - \alpha\beta_1)}{\sqrt{\alpha\beta_1^2}} < -z_\alpha \right) = P\left(\frac{\sqrt{n}(\bar{X} - \alpha\beta_2 + \alpha\beta_2 - \alpha\beta_1)}{\sqrt{\alpha\beta_2^2}} \cdot \frac{\beta_1}{\beta_2} < -z_\alpha \right) = \Phi\left(\frac{\beta_2}{\beta_1} z_\alpha - \frac{\sqrt{n}\alpha_0(\beta_2 - \beta_1)}{\sqrt{\alpha\beta_2^2}} \right)$

  $= \Phi(1.490) = 0.932$

4. Let X be a r.v. distributed as $B(n, \theta)$, $\theta \in \Theta = (0, 1)$.

   - Derive the UMP test for testing the hypothesis $H_0 : \theta \le \theta_0 \longleftrightarrow H_1 : \theta > \theta_0$ at level of significance α.
   - Specify the test in part (i) for $n = 10, \theta_0 = 0.25$, and $\alpha = 0.05$.
   - Compute the power of the test for $\theta = 0.375, 0.500$.
   - Use the CLT to determine the sample size n such that $\theta_0 = 0.125, \alpha = 0.1$ and $\pi(0.25) \ge 0.9$.

- this test is equivalent to X_i ~ i.i.d sample of $B(1, \theta)$

  $f(x) = \theta^x (1-\theta)^x = (1-\theta) e^{\ln\left(\frac{\theta}{1-\theta}\right)x}$ belongs to exponential family

  $Q(\theta) = \ln\frac{\theta}{1-\theta}$ is strictly increasing $\quad T(x) = x \quad V(x) = \sum x_i \sim B(n, \theta)$

  $\varphi(x) = \begin{cases} 1, & x > c \\ r, & x = c \\ 0, & \text{else} \end{cases}$ $\qquad P(X>c) + r\,P(X=c) = \alpha$ is the UMP of the test

– when $n=10$ $\theta_0=0.25$ $\alpha=0.05$

$\Rightarrow c=5$ $\quad r=0.518$ thus $\varphi(x)=\begin{cases}1, & x>5\\ 0.518, & x=5\\ 0, & \text{else}\end{cases}$

– $\pi(\theta)=P(X>c)+rP(X=c)=P(X>5)+0.518P(X=5)$

$\pi(0.375)=0.220$ $\quad \pi(0.500)=0.504$

– By CLT, $\frac{X-n\theta}{\sqrt{n\theta(1-\theta)}}\xrightarrow{d}N(0,1)$ $\quad \varphi(x)=\begin{cases}1, & \frac{X-n\theta_0}{\sqrt{n\theta_0(1-\theta_0)}}>z_\alpha\\ 0, & \end{cases}$

$\pi(\theta)=P\left(\frac{X-n\theta+n\theta-n\theta_0}{\sqrt{n\theta(1-\theta)}}\cdot\sqrt{\frac{\theta(1-\theta)}{\theta_0(1-\theta_0)}}>z_\alpha\right)=1-\Phi\left(\sqrt{\frac{\theta_0(1-\theta_0)}{\theta(1-\theta)}}\cdot z_\alpha-\frac{\sqrt{n}(\theta-\theta_0)}{\sqrt{\theta(1-\theta)}}\right)$

$\pi(0.45)\geq 0.9$ $\quad \theta_0=0.375$ $\quad \alpha=0.1$

$\Rightarrow n=62$

5. Let X be a r.v. with p.d.f. $f(x;\theta)=\frac{1}{\theta}e^{-x/\theta}, x>0, \theta\in\Theta=(0,+\infty)$.

   – Show that $f(\cdot;\theta)$ is an exponential family.
   – Derive the UMP test for testing the hypothesis $H_0:\theta\leq\theta_0 \longleftrightarrow H_1:\theta>\theta_0$ at level of significance α, on the basis of the random sample $X_1,\cdots,X_n$ from the above p.d.f..
   – If $\theta_0=1000$ and $\alpha=0.05$, determine the sample size n such that the power of the test at $\theta=1500$ is at least 0.95.

– $f(x;\theta)=\frac{1}{\theta}e^{-\frac{x}{\theta}}I_{x>0}$ $\quad T(x)=x$ $\quad Q(\theta)=-\frac{1}{\theta}$ strictly increasing

$f(\cdot;\theta)$ is an exponential family

– $V(x)=\sum x_i$ $\quad \varphi(x)=\begin{cases}1, & \sum x_i>c\\ 0, & \text{else}\end{cases}$

$V(x)\sim \text{Gamma}(n,1/\theta)$ $\quad P(V(x)>c)=\alpha \Rightarrow \int_0^c \frac{\theta_0^{-n}}{\Gamma(n)}x^{n-1}e^{-\frac{1}{\theta_0}x}dx=1-\alpha$

– $\pi(\theta)=1-\int_0^c \frac{\theta^{-n}}{\Gamma(n)}x^{n-1}e^{-\frac{1}{\theta}x}dx$

$\therefore$ $\begin{cases}\int_0^c \frac{1500^{-n}}{\Gamma(n)}x^{n-1}e^{-\frac{1}{1500}x}dx=0.05\\ \int_0^c \frac{1000^{-n}}{\Gamma(n)}x^{n-1}e^{-\frac{1}{1000}x}dx=0.05\end{cases}$ $\Rightarrow n=67$

6. Consider the data set "data.csv" in homework 4. For simplicity, suppose that the dependent variable SalePrice (denoted as y_i) and the independent variables Gr_Liv_Area (denoted as x_{i1}), Central_Air (denoted as x_{i2}) follow a linear regression model with an intercept, i.e.,

$$y_i=\beta_0+\beta_1x_{i1}+\beta_2x_{i2}+\epsilon_i,\quad \epsilon_i \text{ i.i.d. } \sim N(0,\sigma^2).$$

Testing the following hypotheses:

   – $H_0:\beta_1=0 \longleftrightarrow H_1:\beta_1\neq 0$.
   – $H_0:\beta_2=0 \longleftrightarrow H_1:\beta_2\neq 0$.
   – $H_0:\beta_1=\beta_2=0 \longleftrightarrow H_1:\beta_1\neq 0$ or $\beta_2\neq 0$.
   – $H_0:\sigma=50000 \longleftrightarrow H_1:\sigma\neq 50000$.

– considering test $H_{0j}:\beta_j=0 \leftrightarrow H_1:\beta_j\neq 0$

$\hat{\beta}_j$ is a good estimator of β_j $\quad D_j=\left\{(\vec{X},\vec{Y}) \,\middle|\, \frac{|\hat{\beta}_j|}{\hat{\sigma}\left[(X^TX)^{-1}\right]^{\frac{1}{2}}_{j+1,j+1}}>y\right\}$

by significance level α, $c = t_{n-p-1,\alpha/2}$

P-value $p = P_{\beta_j=0}(|T_j| > |t_j|) = P(|t_{n-p-1}| > |t_j|)$

using python, with $\alpha = 0.05$

```
Hypothesis Testing with sigificance level  0.05

H0: beta_X1 = 0, p-value = 4.719758197824799e-255
Reject H0: There is a significant relationship between X1 and Y.

H0: beta_X2 = 0, p-value = 4.767979759130875e-24
Reject H0: There is a significant relationship between X2 and Y.
```

— test $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0 \leftrightarrow H_1: \beta_i = 0 \ \exists i \in [1,p]$

$L(\beta, \sigma^2; \vec{X}, \vec{y}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{\|Y - X\beta\|_2^2}{\sigma^2}\right\}$ $\Theta = \{(\vec{\beta}, \sigma^2), \beta \in R^p, \sigma^2 > 0\} \in R^{p+2}$

$L(\Omega) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\{-\frac{n}{2}\}$ $\hat{\beta} = (X^T X)^{-1} X^T Y$ $\hat{\sigma}^2_{MLE} = \frac{1}{n}\|Y - X\hat{\beta}\|_2^2$

under H_0, $\Theta_0 = \{(\vec{\beta}, \sigma^2): \beta = (\beta_0, \vec{0}), \sigma^2 > 0\} \in R^2$

$\hat{\beta}_0 = \bar{y}$ $\hat{\sigma}^2 = \frac{1}{n}\|Y - \bar{y}\|_2^2$ $L(\hat{w}) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\{-\frac{n}{2}\}$

$\lambda(\vec{X}, \vec{y}) = \left(\frac{\|Y - X\hat{\beta}\|_2}{\|Y - \bar{y}\|_2}\right)^n = \left[\frac{\|Y - X\hat{\beta}\|_2^2 + \|X\hat{\beta} - \bar{y}\|_2^2}{\|Y - X\hat{\beta}\|_2^2}\right]^{-\frac{n}{2}}$

$Z_1 = \frac{\|X\hat{\beta} - \bar{y}\|_2^2}{\sigma^2} \sim \chi^2_p$ $Z_2 = \frac{\|Y - X\hat{\beta}\|_2^2}{\sigma^2} \sim \chi^2_{n-p-1}$ $F = \frac{Z_1/p}{Z_2/(n-p-1)} \sim F_{p,n-p-1}$

thus, rejection region $D = \{(\vec{y}, \vec{X}) \mid F > F_{p,n-p-1,\alpha}\}$

```
Joint Test for Beta1 and Beta2:
<F test: F=1002.3955504614278, p=1.1762570531942729e-282, df_denom=1.6e+03, df_num=2>

Hypothesis Testing with sigificance level  0.05

H0: beta1 = beta2 = 0, p-value = 1.1762570531942729e-282
Reject H0: There is a significant relationship between X1 and X2.
```

— $H_0: \sigma^2 = \sigma_0^2 \leftrightarrow H_1: \sigma^2 \neq \sigma_0^2$

under H_0: $\Theta_0 = \{\vec{\beta}\} \in R^p$ $\hat{\beta} = (X^T X)^{-1} X^T Y$ $L(\Omega) = (2\pi\sigma_0^2)^{-\frac{n}{2}} e^{\exp\{-\frac{n}{2}\}}$

$\lambda(\vec{X}, \vec{y}) = \left[\frac{\|Y - X\hat{\beta}\|_2^2}{n\sigma_0^2}\right]^{-\frac{n}{2}}$ $-2\log\lambda \xrightarrow{d} \chi^2_1$

$D = \{(\vec{y}, \vec{X}) \mid |n(\ln\|Y - X\hat{\beta}\|_2^2 - \ln n\sigma_0^2)| > \chi^2_{1,\alpha_2}\}$

```
residuals_sum_of_squares:  4469456563876.83
```

$(79.33) > \chi^2_{1,\alpha/2}$

so we reject H_0

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.557
Model:                            OLS   Adj. R-squared:                  0.556
Method:                 Least Squares   F-statistic:                     1002.
Date:                Fri, 29 Dec 2023   Prob (F-statistic):          1.18e-282
Time:                        23:58:12   Log-Likelihood:                -19647.
No. Observations:                1598   AIC:                         3.930e+04
Df Residuals:                    1595   BIC:                         3.932e+04
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -3.758e+04   5837.338     -6.439      0.000   -4.9e+04   -2.61e+04
X1           113.7358      2.747     41.402      0.000     108.347     119.124
X2          5.173e+04   5031.497     10.280      0.000    4.19e+04    6.16e+04
==============================================================================
Omnibus:                      396.630   Durbin-Watson:                   1.983
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1720.844
Skew:                           1.121   Prob(JB):                         0.00
Kurtosis:                       7.563   Cond. No.                     8.40e+03
==============================================================================
```

code listed next page

7. Apply the two-way ANOVA to the data given in the table below. Take $\alpha = 0.05$.

Levels of Factor A \ Levels of Factor B	1	2	3	4	5	
1	110	128	48	123	19	85.6
2	214	183	115	114	129	151
3	208	183	130	225	114	172
	177.3	164.7	97.7	154	87.3	136.2

$\hat{\alpha}_1 = -50.6 \quad \hat{\alpha}_2 = 14.8 \quad \hat{\alpha}_3 = 35.8$

$\hat{\beta}_1 = \frac{617}{15} \quad \hat{\beta}_2 = \frac{423}{15} \quad \hat{\beta}_3 = -\frac{578}{15} \quad \hat{\beta}_4 = -\frac{733}{15}$

$SS_A = 5 \times [85.6^2 + 151^2 + 172^2] - 15 \times (136.2)^2 = 20303.2$

$SS_B = 3 \times [177.3^2 + 164.7^2 + 97.7^2 + 154^2 + 87.3^2] - 15 \times (136.2)^2 = 20073.73$

$SS_T = 47442.4$

$\therefore SSe = 7061.46$

unbiased σ^2 is $MSe = 882.6833$

$F_A = \frac{MS_A}{MSe} = 11.50 \quad F_B = \frac{MS_B}{MSe} = 5.69$

$F_{2,8,0.05} = 4.459 \quad F_{4,8,0.05} = 3.83$

so we reject H_{0A} & H_{0B}

```python
import numpy as np
import statsmodels.api as sm
import pandas as pd

csv_path = "./data.csv"
data = pd.read_csv(csv_path)

SalePrice = data["SalePrice"]
Gr_Liv_Area = data["Gr_Liv_Area"]
Central_Air = data["Central_Air"]

Central_Air = Central_Air.replace("Y", 1)
Central_Air = Central_Air.replace("N", 0)

X1 = np.array(Gr_Liv_Area)
X2 = np.array(Central_Air)
Y = np.array(SalePrice)

# 构建数据框
data = pd.DataFrame({"X1": X1, "X2": X2, "Y": Y})

# 建立多参数线性回归模型
X = sm.add_constant(data[["X1", "X2"]])
 # 添加常数项
model = sm.OLS(data["Y"], X).fit()

# 输出回归结果
print(model.summary())

# 获取回归系数和p值
coefficients = model.params
p_values = model.pvalues

# 计算残差平方和
residuals = model.resid
residuals_sum_of_squares =
np.sum(np.square(residuals))
print("residuals_sum_of_squares: ",
residuals_sum_of_squares)

# 输出回归系数和p值
print("\nCoefficients:")
print(coefficients)
print("\nP-values:")
print(p_values)

# 进行假设检验
alpha = 0.05
print("\nHypothesis Testing with
sigificance level ", alpha)
print(f"\nH0: beta_X1 = 0, p-value =
{p_values['X1']}")
if p_values["X1"] < alpha:
    print("Reject H0: There is a
significant relationship between X1 and
Y.")
else:
    print("Fail to reject H0: There is no
significant relationship between X1 and
Y.")

print(f"\nH0: beta_X2 = 0, p-value =
{p_values['X2']}")
if p_values["X2"] < alpha:
    print("Reject H0: There is a
significant relationship between X2 and
Y.")
else:
    print("Fail to reject H0: There is no
significant relationship between X2 and
Y.")

# 假设检验：H0: beta1 = beta2 = 0
joint_test = model.f_test("X1 = X2 = 0")

# 输出检验结果
print("\nJoint Test for Beta1 and
Beta2:")
print(joint_test)

# 获取检验结果的p值
p_value = joint_test.pvalue

# 进行假设检验
print("\nHypothesis Testing with
sigificance level ", alpha)
print(f"\nH0: beta1 = beta2 = 0, p-value
= {p_value}")
if p_value < alpha:
    print("Reject H0: There is a
significant relationship between X1 and
X2.")
else:
    print("Fail to reject H0: There is no
significant relationship between X1 and
X2.")
```